Properly colored paths in the union of two Hamiltonian paths: the $O(\sqrt{n}\,)$ collapse, the transit floor, and the failure of block lower bounds

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Abstract

Overlay two Hamiltonian paths on a common vertex set of size n, one blue and one red. How long a properly colored path – consecutive edges alternating in color – must the union contain? Every such union is H_σ for a permutation σ of $\{0,\ldots,n-1\}$, with blue edges joining consecutive positions and red edges consecutive values; write $\rho(\sigma)$ for the maximum number of vertices on a properly colored path in H_σ, and $\rho_{\min}(n) = \min_\sigma \rho(\sigma)$. We prove $\rho_{\min}(n) = O(\sqrt{n}\,)$, explicitly $\rho_{\min}(n) \le 8\sqrt{n} + 20$ for $n \ge 100$, by inflating a gadget whose *transit number* equals 3 – and we prove that 3 is optimal for every gadget size. In the other direction we show that block statistics yield no lower bound: there are *block-free* permutations ($b(\sigma) = n$ maximal ± 1-blocks) with $\rho = \Theta(\sqrt{n}\,)$, so $\inf_\sigma \rho(\sigma)/b(\sigma) = 0$, and the inequality $\rho \ge b(\sigma)$, which holds exhaustively through $n = 12$, first fails at $n = 14$. The problem the paper poses is the lower bound: *does* $\rho_{\min}(n) \to \infty$*?* Two overlaid spanning paths look far too connected for the answer to be no, yet no lower bound growing with n is known, and our results show that any such bound must stop at $O(\sqrt{n}\,)$.

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1 Introduction

Let S_n denote the set of bijections $\sigma\colon \{0,1,\ldots,n-1\} \to \{0,1,\ldots,n-1\}$. Associate with $\sigma \in S_n$ the two-edge-colored multigraph H_σ on vertex set $\{0,1,\ldots,n-1\}$ with

- **P-edges** ("position", blue): $\{i, i+1\}$ for $0 \le i \le n-2$;

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- **V-edges** ("value", red): $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ for $0 \le v \le n-2$.

Each color class is a Hamiltonian path. A vertex pair may carry both a P-edge and a V-edge; these are retained as *parallel edges of different colors* ("doubled pairs"). Conversely, the union of any two Hamiltonian paths on a common vertex set, with one path blue and the other red, is color-isomorphic to H_σ for some σ (identify the vertex set with $\{0, \ldots, n-1\}$ along the blue path, and let σ rank the vertices along the red path).

A *properly colored path* (PC path) is a simple path—all vertices distinct—whose consecutive edges have different colors. A single vertex is a PC path. Let $\rho(\sigma)$ be the maximum number of vertices on a PC path in H_σ, and

$$\rho_{\min}(n) \;=\; \min_{\sigma \in S_n} \rho(\sigma).$$

A useful picture. Plot σ as the point set $\{(i, \sigma(i)) : 0 \le i \le n-1\}$ in the plane. A P-edge joins horizontally adjacent columns; a V-edge joins vertically adjacent rows. A PC path traces an alternating horizontal/vertical staircase through this grid. Both color classes have maximum degree 2, so H_σ has maximum degree 4; this places the problem outside the regimes of standard properly colored path results, which typically require minimum degree linear in n [6].

1.1 Motivation: a relaxation of a matching-edge parameter

Lovász [1] asked in 1969 whether every connected vertex-transitive graph on n vertices contains a Hamiltonian path. The best general bound is $\Omega(n^{9/14})$, due to Norin, Steiner, Thomassé and Wollan [2], and rests on a reduction that depends on the matching-edge content of long simple paths in a family of graphs $G(n, \sigma)$ on $2n$ vertices, indexed by $\sigma \in S_n$. Writing

$$f(n) \;=\; \min_{\sigma \in S_n} \; \max_{\pi} \; \mu(\pi),$$

where π ranges over simple paths in $G(n, \sigma)$ and $\mu(\pi)$ counts the matching edges of π, the current bound $f(n) = \Omega(n^{4/5})$ comes from the circumference theorem of Liu, Yu and Zhang [3] on 3-connected cubic graphs, as used in [2]; replacing it with $f(n) = \Omega(n)$ would yield $\Omega(n^{2/3})$ on the Lovász problem. We define $G(n, \sigma)$ precisely in Section 3 and prove there a one-directional *conversion lemma*: a PC path on k vertices in H_σ yields a simple path in $G(n, \sigma)$ using k matching edges, so $f(n) \ge \rho_{\min}(n)$. A linear lower bound on $\rho_{\min}$ would thus give a linear lower bound on f. The relaxation in this direction is the route we introduce, and the main theorem closes it. The scope of that statement is delimited in Section 1.3.

Beyond this specific motivation, the question is a natural extremal problem about alternating paths. Properly colored ("alternating") paths and cycles in edge-colored graphs have been studied extensively since the survey of Bang-Jensen and Gutin [4]; see also [5, 6]. The positive results in that line are sufficient conditions—typically large color degree or dense hosts—for long PC paths or cycles to exist. The graphs H_σ sit at the opposite end of the spectrum: every vertex has at most two edges of each color, and the question is extremal rather than Dirac-type. We are aware of no prior work on the union of two Hamiltonian paths in this guise.

1.2 Results

The clean self-contained question, for a reader who sets the Lovász application aside, is:

Does there exist $c > 0$ such that every two-edge-colored multigraph that is the union of two Hamiltonian paths on the same vertex set contains a properly colored simple path on at least cn vertices?

We answer it in the negative, and quantitatively.

Theorem 1.1 (Main Theorem)**.** *$\rho_{\min}(n) = O(\sqrt{n})$. Explicitly, $\rho_{\min}(n) \le 8\sqrt{n} + 20$ for all $n \ge 100$. Consequently $\rho_{\min}(n)/n \to 0$.*

The question we consider central, and cannot answer, points the other way:

Does $\rho_{\min}(n) \to \infty$?

A union of two Hamiltonian paths is spanning and connected, and every vertex sits on two long monochromatic paths; it is hard to believe that its longest properly colored path could stay *bounded* as $n \to \infty$. Nothing we know rules it out. The two calibration results of this paper quantify part of the difficulty: the transit floor (Section 9) shows the construction machinery cannot be improved by better gadgets, and the block-count refutations (Section 10) close the most natural lower-bound route while capping any future lower bound at $O(\sqrt{n})$. We hope the question finds attackers.

The route to Theorem 1.1 passes through a refutation. Small cases and a locally extremal family pointed at a linear lower bound with constant 1/2:

Half-bound (refuted, Section 4): for every $\sigma \in S_n$, $\rho(\sigma) \ge \lceil (n+3)/2 \rceil$.

This holds on a block-reversal family σ_m (which attains it exactly) and through exhaustive enumeration of S_n for $n \le 12$, but it is false: at $n = 18$ an explicit permutation σ^* has $\rho(\sigma^*) = 10 < 11$ (Theorem 4.6). The construction of σ^* – deleting one singleton from σ_4 to fuse two 3-blocks – is the seed of the gadget idea, and the main theorem shows that the half-bound fails not by an additive constant but by a polynomial factor.

The construction behind the main theorem is explicit. For even $b \ge 6$ let

$$T_b \;=\; (0,\; b{-}1,\; 2,\; 3,\; \ldots,\; b{-}2,\; 1) \in S_b$$

be the identity permutation with the values 1 and $b{-}1$ exchanged, and for $a \ge 2$ define $\sigma_{a,b} \in S_{ab}$ by

$$\sigma_{a,b}(jb + r) \;=\; jb + T_b(r) \qquad (0 \le j < a,\; 0 \le r < b),$$

i.e. a consecutive copies of T_b on consecutive value ranges. We prove the two-sided estimate

$$2a + 2b - 7 \;\le\; \rho(\sigma_{a,b}) \;\le\; 4a + 2b \qquad (a \ge 2,\; b \ge 6 \text{ even}), \tag{1}$$

the upper bound by the inflation calculus below (Theorem 8.1), the lower bound by an explicit path (Proposition 8.3). Thus $\rho(\sigma_{a,b}) = \Theta(a + b)$: with $a \asymp b \asymp \sqrt{n}$ the family realizes the order $\sqrt{n}$ exactly, so the upper bound is tight for this family and not an artifact of a construction whose true value is smaller. (For $a = 2$ both bounds in (1) are weak; the construction is informative for $a \ge 3$, and asymptotically for $a \to \infty$.) Exact machine computation gives $\rho(\sigma_{a,b}) = 2a + 2b - 4$ at every tested point up to $n = 720$ (Section 11); we use this exact formula nowhere in the proofs and state it only as data.

The proof of the upper bounds has two independent components.

1. An inflation calculus (Sections 5–6). Call a set B of consecutive positions whose σ-image is a set of consecutive values an *interval block*. A short argument (Lemma 5.2) shows that at most *four* edges of H_σ join B to its complement, and that they attach at four canonical *terminal* vertices of B, with prescribed colors. If $\sigma = \tau[\pi_1, \ldots, \pi_a]$ is built by inflation (each position block an interval block with pattern π_i, arranged by an outer permutation τ), then any PC path decomposes into at most $2a-1$ *visits* to blocks, and each visit other than the first and last must enter and leave through terminals subject to color constraints. Defining the *transit number* $M(\pi)$ as the largest number of vertices such a constrained passage (*admissible visit*) through H_π can collect, we obtain (Theorem 6.4)

$$\rho\big(\tau[\pi_1, \ldots, \pi_a]\big) \;\le\; (2a-3)\,\max_i M(\pi_i) \;+\; 2\,\max_i |\pi_i|.$$

2. A classification of the gadget's admissible visits (Section 7). For every even $b \ge 6$ we determine the admissible visits of H_{T_b} *completely* (Theorem 7.4): up to reversal there are exactly five, of sizes $1, 2, 2, 3, 3$, so $M(T_b) = 3$, independent of b. The mechanism is a parity obstruction: H_{T_b} contains a long run of doubled edges which any terminal-to-terminal passage must either avoid or cross completely, and a full crossing forces the color of the departing edge precisely so that, for even b, every continuation is blocked. For odd b the parity reverses and a transit can collect all b vertices; exact computation confirms $M(T_b) = b$ for odd b. The even/odd contrast shows that the parity, not some feature of the argument, is what bounds the transit number.

Combining the two components and balancing a against b yields Theorem 1.1 (proved as Theorem 8.1). The two-visit refinement of the classification (Corollary 7.7) gives the stated constant $8\sqrt{n} + 20$; the value $M(T_b) = 3$ alone, without the full classification, already suffices for the weaker $\rho_{\min}(n) \le 10\sqrt{n} + 10$ (Remark 8.2).

Appendix A contains a *second, independent* construction proving $\rho_{\min}(n) = O(\sqrt{n})$: chains of reversed odd blocks separated by singletons, where a different parity trap blocks every crossing, with its own complete proof. The two constructions share the parity-trap idea but are different families with different edge inventories, and they were found and verified independently, by different methods and different solvers. We give both because a redundant proof of the main claim is worth the space.

The paper then proves two calibration results, one on each side of the main theorem.

3. The transit floor (Section 9). The inflation calculus makes one parameter decisive: how small a transit number a pattern can have. We prove that the gadgets above are optimal in this parameter:

$$\mu(b) \;:=\; \min_{\pi \in S_b} M(\pi) \;=\; 3 \qquad \text{for every } b \ge 3.$$

The lower bound $M(\pi) \ge 3$ for *every* pattern is proved by an elementary matching-union argument: choosing one matching inside each color's Hamiltonian path with prescribed exposed vertices forces their union to contain a properly colored terminal-to-terminal path that is itself an admissible visit of size at least 3. The upper bound is T_b for even b together with a sibling gadget T'_b (the identity with values 2 and $b-1$ exchanged) for odd b, whose admissible visits we also classify completely (Appendix B). Consequently no choice of inflation blocks can improve the visit-count bookkeeping of the method by more than constants.

4. Block counts do not bound ρ (Section 10). Writing $b(\sigma)$ for the number of maximal ± 1-blocks, the inequality $\rho(\sigma) \geq c\, b(\sigma)$ for any fixed $c > 0$ would imply $\rho_{\min}(n) \geq c\sqrt{n}$ (Section 12), and $\rho \geq b(\sigma)$ holds exhaustively through $n = 12$. We show this route is closed: there are *block-free* permutations ($b(\sigma) = n$) built as chains of a modular staircase pattern whose longest properly colored path has $\Theta(\sqrt{n})$ vertices, so $\inf_\sigma \rho(\sigma)/b(\sigma) = 0$. The proof is a visit analysis in the inflation framework of Section 6: a bipartite parity obstruction pins every through-transit of a chain piece to a *single vertex*, so all but four pieces of the chain contribute at most 2 vertices each. The same computation that locates counterexamples finds the exact threshold for the case $c = 1$: the smallest counterexamples to $\rho \geq b(\sigma)$ occur at $n = 14$ (an explicit block-free witness with $\rho = 13$; exactly 16 such permutations at $n = 14$, none for block-free $n \leq 13$ nor for arbitrary σ with $n \leq 12$). The first counterexamples we had found lay at $n = 240$, inside the family $\sigma_{a,b}$; the true threshold is 14.

Section 12 collects what remains of the lower-bound question – which is now known to require tools beyond block statistics: even $\rho_{\min}(n) \to \infty$ is open, and by Section 10 a lower bound of order $\sqrt{n}$ would be best possible.

1.3 Scope: what this does and does not say about Lovász's problem

The conversion lemma of Section 3 maps PC paths in H_σ to matching-edge-rich paths in $G(n, \sigma)$ *one-directionally*. Our construction makes properly colored paths short; it says nothing about paths in $G(n, \sigma)$ that are allowed to use runs of same-colored edges. The known bound $f(n) = \Omega(n^{4/5})$, which rests on the circumference theorem of [3] as used in [2], is untouched, and $f(n) = \Theta(n)$ remains entirely possible. Thus the construction rules out only the approach of proving $f(n) = \Omega(n)$ via the properly colored relaxation introduced here.

On the lower-bound side we prove no superconstant lower bound on $\rho_{\min}(n)$, and to our knowledge even $\rho_{\min}(n) \to \infty$ is open. What this paper adds on that side is negative but sharp: the most natural route to a lower bound – forcing ρ to grow with the number of ± 1-blocks – is provably closed (Section 10), and any lower bound of order $\sqrt{n}$ would be best possible. The remaining prospects are discussed in Section 12, where we also state the conjecture $\rho_{\min}(n) = \Theta(\sqrt{n})$ – at present an upper bound plus structural circumstantial evidence, no more.

2 Preliminaries

Throughout, $\sigma \in S_n$ and H_σ are as in Section 1; colors are P and V, and for a color c we write $\bar{c}$ for the other color. Paths are sequences of distinct vertices $u_1, \ldots, u_k$ ($k \geq 1$) together with, for each consecutive pair, a chosen edge (with its color) joining them; in a doubled pair the path commits to one of the two parallel edges. A path is *properly colored* (PC) if consecutive edges have different colors. The *size* of a path is its number of vertices k; a PC path on k vertices has $k - 1$ edges. Since each vertex pair carries at most one edge of each color and paths have distinct vertices, a path uses pairwise distinct edges, even in the multigraph sense.

Observation 2.1. A two-edge-colored multigraph is the union of a blue and a red Hamiltonian path on a common vertex set of size n if and only if it is color-isomorphic to H_σ for some $\sigma \in S_n$. Hence Theorem 1.1 is equivalent to its abstract formulation.

Remark 2.2 (Two notions of block)**.** Two block notions recur, and we keep them distinct. An *interval block* (Definition 5.1) is any set of consecutive positions whose values are consecutive; it carries an arbitrary *pattern*, and is the unit of the inflation calculus. A ±1-*block* (Definition 4.1) is the special case in which consecutive values differ by a fixed ±1; these are intrinsic to σ, partition the positions, and are the unit of the lower-bound discussion (Section 12). Every ±1-block is an interval block; the converse fails.

3 The conversion lemma

This section makes precise the link to the matching-edge parameter of Section 1.1. It is logically independent of the rest of the paper and may be skipped by a reader interested only in $\rho_{\min}(n)$.

Definition 3.1. For $\sigma \in S_n$, let $G(n,\sigma)$ be the graph on $2n$ vertices $\{v_0,\dots,v_{n-1},u_0,\dots,u_{n-1}\}$ with edges: a *top path* $v_0v_1\cdots v_{n-1}$; a *bottom path* $u_0u_1\cdots u_{n-1}$; and a *matching* $M = \{v_iu_{\sigma(i)} : 0 \le i \le n-1\}$. Every v_i and every u_j is incident to exactly one matching edge; the four path endpoints have degree 2 and every other vertex degree 3. For a simple path π in $G(n,\sigma)$, $\mu(\pi)$ is the number of edges of π in M, and $f(n) = \min_\sigma \max_\pi \mu(\pi)$.

Theorem 3.2 (Conversion Lemma)**.** *If H_σ has a PC path on k vertices, then $G(n,\sigma)$ has a simple path on $2k$ vertices using exactly k matching edges. Hence* $\max_\pi \mu(\pi) \ge \rho(\sigma)$ *for every σ, and $f(n) \ge \rho_{\min}(n)$.*

Proof. Let $w_0w_1\cdots w_{k-1}$ be a PC path in H_σ with edge colors $c_0,\dots,c_{k-2}$ alternating. Build a path Π in $G(n,\sigma)$ on the $2k$ vertices $\{v_{w_i}\} \cup \{u_{\sigma(w_i)}\}$ as follows. If $c_0 = \mathrm{P}$, run

$$u_{\sigma(w_0)} \xrightarrow{M} v_{w_0} \xrightarrow{P} v_{w_1} \xrightarrow{M} u_{\sigma(w_1)} \xrightarrow{Q} u_{\sigma(w_2)} \xrightarrow{M} v_{w_2} \xrightarrow{P} v_{w_3} \xrightarrow{M} \cdots$$

alternating matching edges with top/bottom path edges; if $c_0 = \mathrm{V}$, start instead at v_{w_0}, symmetrically.

Edge validity. A P-edge $\{w_i, w_{i+1}\}$ means $|w_i - w_{i+1}| = 1$, so $v_{w_i}v_{w_{i+1}}$ is a top-path edge; a V-edge $\{w_i, w_{i+1}\}$ means $|\sigma(w_i) - \sigma(w_{i+1})| = 1$, so $u_{\sigma(w_i)}u_{\sigma(w_{i+1})}$ is a bottom-path edge. The color alternation of the PC path matches the alternation of top versus bottom traversal in Π, so every step of Π is a genuine edge.

Simplicity. The top vertices $\{v_{w_i}\}$ are distinct because the w_i are; the bottom vertices $\{u_{\sigma(w_i)}\}$ are distinct because σ is a bijection; the two families are disjoint.

Matching count. Π uses one matching edge per vertex of the PC path, namely $v_{w_i}u_{\sigma(w_i)}$ for $0 \le i < k$: a total of k.

Taking $\pi = \Pi$ gives $\max_\pi \mu(\pi) \ge \rho(\sigma)$ for each σ; the minimum over σ gives $f(n) \ge \rho_{\min}(n)$. □

The conversion is one-directional: a path in $G(n,\sigma)$ that uses a run of two same-colored path edges (top or bottom) has no PC-path preimage, so the inequality $f(n) \ge \rho_{\min}(n)$ cannot be reversed. This is the precise sense in which our negative result for $\rho_{\min}$ does not transfer to f (Section 1.3).

4 The half-bound and its refutation

This section records the false start that motivates the rest of the paper. We exhibit a family on which ρ equals $\lceil (n+3)/2 \rceil$, state the half-bound it suggested, and then refute that half-bound by an explicit permutation at $n = 18$. The lower-bound construction and the refutation are proved in full; the exact value of the family is stated as computed data and is not used anywhere downstream.

Definition 4.1 (± 1-block)**.** A ± 1-*block* of σ is a maximal interval of positions $[p_\ell, p_r]$ such that either $p_\ell = p_r$ (a *singleton*) or there is a fixed $\varepsilon \in \{+1, -1\}$ with $\sigma(i+1) - \sigma(i) = \varepsilon$ for all $p_\ell \le i < p_r$. The ± 1-blocks partition $\{0, \ldots, n-1\}$; write $b(\sigma)$ for their number and $s_{\max}(\sigma)$ for the largest size.

Every internal edge of a ± 1-block is doubled (consecutive positions hold consecutive values), so a ± 1-block can be traversed in order with freely chosen alternating colors. This single fact gives two unconditional bounds.

Proposition 4.2 (Largest-block and one-vertex bounds)**.** *For every $\sigma \in S_n$, $\rho(\sigma) \ge s_{\max}(\sigma)$; and if $b(\sigma) \ge 2$, then $\rho(\sigma) \ge s_{\max}(\sigma) + 1$.*

Proof. Let B be a ± 1-block of size $s_{\max}$. Its internal edges are doubled, so traversing B in order with alternating $\mathrm{P}, \mathrm{V}, \ldots$ is a PC path on $s_{\max}$ vertices: this gives the first bound.

For the second, if $s_{\max} = 1$ then all blocks are singletons; with $b \ge 2$ the P-edge $\{0, 1\}$ exists and is a PC path on $2 = s_{\max} + 1$ vertices. Otherwise $s_{\max} \ge 2$. Since $b \ge 2$, a largest block B has a ± 1-block neighbor, hence an external edge e of some color c at a port $p \in \{p_\ell, p_r\}$ (Lemma 5.2 below gives the port structure for interval blocks, of which B is one). Traverse B from the port opposite p toward p, choosing the starting color so that the last internal edge has color $\neq c$ (the doubled internal edges allow either parity); then e, of color c, may be appended at p, yielding a PC path on $s_{\max} + 1$ vertices. ☐

These are weak in absolute terms – the gadget family below has $s_{\max} = b$ fixed while $n \to \infty$ – but they are the floor any combined bound must clear, and they are the only unconditional lower bounds we have (Section 12).

4.1 The block-reversal family and the half-bound

Definition 4.3. For $m \ge 1$ and $n = 4m + 3$, partition $\{0, \ldots, n-1\}$ into consecutive blocks of sizes

$$\big(2,\ \underbrace{3, 1, 3, 1, \ldots, 3, 1, 3}_{m \text{ threes, } m-1 \text{ singletons}},\ 2\big),$$

and let σ_m be the permutation that reverses each block (the values of each block run downward, with consecutive blocks carrying consecutive value ranges). For $m = 1$ this is the composition $(2, 3, 2)$ with no interior singleton. Explicitly,

$$\begin{aligned}
\sigma_1 &= (1, 0, 4, 3, 2, 6, 5), \\
\sigma_2 &= (1, 0, 4, 3, 2, 5, 8, 7, 6, 10, 9), \\
\sigma_3 &= (1, 0, 4, 3, 2, 5, 8, 7, 6, 9, 12, 11, 10, 14, 13), \\
\sigma_4 &= (1, 0, 4, 3, 2, 5, 8, 7, 6, 9, 12, 11, 10, 13, 16, 15, 14, 18, 17).
\end{aligned}$$

Each block of σ_m is an interval block (its values are consecutive), so σ_m is an inflation in the sense of Section 6; this is the viewpoint that the gadget construction later sharpens.

Proposition 4.4 (An explicit long path)**.** *For every $m \ge 1$, $\rho(\sigma_m) \ge 2m + 3 = (n+3)/2$.*

Proof. Define a vertex sequence Γ_m by starting with $(4m-2,\ 4m-1,\ 4m)$; then for $j = m, m-1, \ldots, 2$ appending the pair $(4j-3,\ 4j-4)$; and finally appending $(0, 1)$. For $m = 4$ this is $\Gamma_4 = (14, 15, 16, 13, 12, 9, 8, 5, 4, 0, 1)$, of length $3 + 2(m-1) + 2 = 2m + 3$. We check Γ_m is a PC path in H_{σ_m}.

The edges fall into four groups, whose colors are forced by Definition 4.3 (direct from the position/value adjacencies):

- the internal edges $(4m-2, 4m-1)$, $(4m-1, 4m)$ and $(0, 1)$ are doubled (each lies inside a ± 1-block), hence freely colorable;
- each edge $(4j, 4j-3)$, $j = m, \ldots, 2$, joins the top of a 3-block to the adjacent singleton across a value-gap of 1 but a position-gap of 3: V-only;
- each edge $(4j-3, 4j-4)$, $j = m, \ldots, 2$, joins a singleton to the next 3-block across a position-gap of 1: P-only;
- the final edge $(4, 0)$ joins the leftmost 3-block to the leftmost 2-block: V-only.

Reading Γ_m in order, the forced colors are $\mathrm{V}, \mathrm{P}, \mathrm{V}, \mathrm{P}, \ldots, \mathrm{V}, \mathrm{P}$ on the $2m+2$ edges, the doubled internal edges taking whichever color the alternation requires. All vertices are distinct, so Γ_m is a PC path on $2m + 3$ vertices. □

Exhaustive computation through $n \le 12$ (the range documented in the ancillary artifact) found $\rho_{\min}(n) = \lceil (n+3)/2 \rceil$ tight at $n = 7, 8, 11, 12$ and slack by 1 elsewhere, with σ_m realizing the bound at $n = 4m + 3$. Together with Proposition 4.4 this suggested:

Conjecture 4.5 (Half-bound; *false*)**.** *For every $n \ge 2$ and every $\sigma \in S_n$, $\rho(\sigma) \ge \lceil (n+3)/2 \rceil$.*

Exact computation gives $\rho(\sigma_m) = 2m + 3$ (so the lower bound of Proposition 4.4 is exact), verified for $m = 1, 2, 3, 4$ by independent solvers (Section 11); we record this as data, not as a theorem, and use it nowhere below. The reason we do not prove the matching upper bound $\rho(\sigma_m) \le 2m + 3$ here is deliberate: the inflation calculus of Section 6 gives only the weaker $\rho(\sigma_m) \le O(m)$ with a larger constant (each 3-block has transit number 3, so the generic bound does not see the singleton-flanking rigidity that pins the value to $2m + 3$), and the sharper argument would expand the paper without serving the main theorem, which makes the family entirely obsolete in Section 7. The role of σ_m here is only to motivate Conjecture 4.5 – and to be refuted.

4.2 An $n = 18$ counterexample

Theorem 4.6. *Let $\sigma^* = (1, 0, 4, 3, 2, 5, 8, 7, 6, 11, 10, 9, 12, 15, 14, 13, 17, 16) \in S_{18}$. Then $\rho(\sigma^*) = 10 < 11 = \lceil (18+3)/2 \rceil$, so Conjecture 4.5 is false at $n = 18$.*

The ± 1-block structure of σ^* has composition $(2, 3, 1, 3, 3, 1, 3, 2)$, i.e.

$$B_1 = \{0, 1\}, \quad B_2 = \{2, 3, 4\}, \quad B_3 = \{5\}, \quad B_4 = \{6, 7, 8\},$$
$$B_5 = \{9, 10, 11\}, \quad B_6 = \{12\}, \quad B_7 = \{13, 14, 15\}, \quad B_8 = \{16, 17\}.$$

Proof. Lower bound. The sequence

$$\Pi^* = (13, 14, 15, 12, 11, 6, 5, 2, 1, 0)$$

is a PC path on 10 vertices in H_{σ^*}. Reading the edges from the definition of σ^*:

$\{13, 14\}$ doubled	$\{14, 15\}$ doubled	$\{12, 15\}$ V-only	$\{11, 12\}$ P-only
$\{6, 11\}$ V-only	$\{5, 6\}$ P-only	$\{2, 5\}$ V-only	$\{1, 2\}$ P-only
$\{0, 1\}$ doubled			

(the V-only edges join positions whose σ^*-values are consecutive but whose positions are not, e.g. $\sigma^{*-1}(8) = 6$, $\sigma^{*-1}(9) = 11$ for $\{6, 11\}$). Assigning the colors V, P, V, P, V, P, V, P, V to the nine edges (the doubled edges take the colors the alternation requires) gives a valid PC traversal, so $\rho(\sigma^*) \geq 10$.

Upper bound. An exact longest-PC-path search (state space: triples of visited set, current vertex, last color; every reachable state visited once, no pruning – Section 11) returns $\rho(\sigma^*) = 10$. Two independently written solvers (Python and C++) agree on the value, and the search at $n = 18$ for this single instance is small. Since $\lceil (18 + 3)/2 \rceil = 11$, the half-bound fails. □

Remark 4.7 (How σ^* was found, and why it matters)**.** σ^* is a single-position deletion from $\sigma_4 \in S_{19}$: delete the center singleton (position 9, value 9) and relabel the values above it down by 1. Removing the singleton fuses the two flanking 3-blocks across a non-± 1 gap, changing the composition from $(2, 3, 1, 3, 1, 3, 1, 3, 2)$ to $(2, 3, 1, 3, 3, 1, 3, 2)$ and dropping $b(\sigma)$ from 9 to 8, while the deficit $d = n - \rho$ stays at 8. A *fused* pair of reversed 3-blocks across a non-± 1 gap is exactly a small instance of the parity trap that the gadget T_b exploits at scale: the deletion that breaks the half-bound is the seed of Theorem 1.1. (The values $n = 15, 16, 17$ have not been checked exhaustively, so we do not claim $n = 18$ is the smallest counterexample value.)

4.3 Structural results that survive

Besides the unconditional Proposition 4.2, one conditional comparison survives the refutation and is used nowhere below but is recorded for completeness. For σ with ± 1-blocks $B_1, \ldots, B_{b(\sigma)}$ in position order, the *block-quotient permutation* τ_σ ranks the blocks by their value range; H_{τ_σ} is the corresponding graph on $b(\sigma)$ vertices.

Proposition 4.8 (Odd-only quotient lift)**.** *If every ± 1-block of σ that is neither first nor last in position order has odd size, then $\rho(\sigma) \geq \rho(\tau_\sigma)$.*

Proof. Take a maximum PC path Q in H_{τ_σ} and lift it to H_σ, replacing each block-vertex visit by a visit to the corresponding block. A block visited as a through-vertex of Q is entered and left by edges of different colors (the PC condition at the block-vertex). The entry and exit cross edges attach at terminals of the block, whose port colors equal the entry and exit colors; two cases arise.

If those two terminals are the *same* physical vertex of the block (a ± 1-block can have a P-terminal and a V-terminal coinciding), lift the quotient vertex to that single vertex: it carries two roles of different port colors, so it forms an admissible visit of type (ii) and the two cross edges meet the PC condition there.

If the two terminals are *distinct* vertices, lift to a full traversal of the interval block, which has size s. The $s - 1$ internal edges alternate, so the entry and exit colors differ if and only if $s - 1$ is even, i.e. s is odd. By hypothesis every internal block has odd size, so each such full traversal meets the PC condition at both ports. (End blocks carry only one external edge and impose no parity constraint; lift them by any traversal reaching their single port.)

Either way the lift uses at least one vertex per block-vertex of Q and respects the PC condition at every junction, so it is a PC path on at least $\rho(\tau_\sigma)$ vertices. □

The hypothesis is met by σ_m (internal blocks of sizes 3 and 1). The naive lift fails for general σ: a quotient path demanding a full traversal of an *even* internal block forces equal entry/exit colors. The unconditional inequality $\rho(\sigma) \geq \rho(\tau_\sigma)$, verified empirically for $n \leq 11$, is open.

5 Interval blocks and ports

We now develop the machinery for the main theorem. The unit is the interval block of Remark 2.2.

Definition 5.1 (Interval block, pattern)**.** An *interval block* of $\sigma \in S_n$ is a set of positions $B = \{p_\ell, p_\ell + 1, \ldots, p_r\}$ whose image is a set of consecutive values $\sigma(B) = \{w_\ell, \ldots, w_r\}$. Its *pattern* is the permutation $\pi \in S_b$, $b = |B|$, defined by $\pi(t) = \sigma(p_\ell + t) - w_\ell$.

Lemma 5.2 (Port structure)**.** *Let B be an interval block of σ with positions $[p_\ell, p_r]$, values $[w_\ell, w_r]$, and pattern π. Then:*

(a) The sub-multigraph of H_σ induced on B is color-isomorphic to H_π under the map $p \mapsto p - p_\ell$.

(b) Every edge of H_σ with exactly one endpoint in B is one of at most four: the P-edges $\{p_\ell - 1, p_\ell\}$ and $\{p_r, p_r + 1\}$ (when they exist), whose endpoints in B are p_ℓ and p_r; and the V-edges $\{\sigma^{-1}(w_\ell - 1), \sigma^{-1}(w_\ell)\}$ and $\{\sigma^{-1}(w_r), \sigma^{-1}(w_r + 1)\}$ (when they exist), whose endpoints in B are $\sigma^{-1}(w_\ell)$ and $\sigma^{-1}(w_r)$.

Proof. (a) P-edges internal to B are the pairs $\{p, p+1\} \subseteq [p_\ell, p_r]$, which correspond exactly to the P-edges of H_π. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ has both endpoints in B iff both $v, v+1 \in [w_\ell, w_r]$ (as $\sigma(B) = [w_\ell, w_r]$ and σ is a bijection), and these correspond exactly to the V-edges of H_π under the stated map, since $\pi^{-1}(v - w_\ell) = \sigma^{-1}(v) - p_\ell$ for $v \in [w_\ell, w_r]$. Doubled pairs match doubled pairs, so the isomorphism preserves colors.

(b) A P-edge $\{p, p+1\}$ with exactly one endpoint in $[p_\ell, p_r]$ satisfies $p + 1 = p_\ell$ or $p = p_r$. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ with exactly one endpoint in B has exactly one of $v, v+1$ in $[w_\ell, w_r]$, forcing $v + 1 = w_\ell$ or $v = w_r$. □

Definition 5.3 (Terminals, roles, port colors)**.** Let $\pi \in S_b$, $b \geq 2$. The *terminals* of H_π are the four *roles*

$$p_0 = \text{vertex } 0, \quad p_1 = \text{vertex } b-1, \quad v_0 = \text{vertex } \pi^{-1}(0), \quad v_1 = \text{vertex } \pi^{-1}(b-1),$$

with *port colors* $\mathrm{pc}(p_0) = \mathrm{pc}(p_1) = \mathrm{P}$ and $\mathrm{pc}(v_0) = \mathrm{pc}(v_1) = \mathrm{V}$. Distinct roles may occupy the same vertex (a role is a label, not a vertex); since $b \geq 2$, two roles with the *same* port color always occupy distinct vertices.

By Lemma 5.2, in any interval block with pattern π the at most four outgoing edges attach (inside the block) precisely at the terminals, and the color of an outgoing edge equals the port color of the role at which it attaches.

Definition 5.4 (Admissible visit, transit number)**.** Let $\pi \in S_b$, $b \geq 2$. An *admissible visit* of H_π is either

(i) a PC path $u_1, \ldots, u_k$ in H_π with $k \geq 2$, together with a choice of roles r at u_1 and r' at u_k, such that the first edge has color $\neq \mathrm{pc}(r)$ and the last edge has color $\neq \mathrm{pc}(r')$; or

(ii) a single vertex carrying two distinct roles r, r' with $\mathrm{pc}(r) \neq \mathrm{pc}(r')$.

The *transit number* $M(\pi)$ is the maximum size of an admissible visit, and $M(\pi) = 0$ if none exists. For $b \leq 1$ set $M(\pi) = 1$.

The color constraints encode that a visit is entered and left by external edges of the roles' port colors (which differ from the first and last internal edges by the PC condition). Two trivial but load-bearing facts:

Lemma 5.5 (Reversal; endpoint constraint)**.** (a) *If $(u_1, \ldots, u_k; r, r')$ is an admissible visit, so is the reversed $(u_k, \ldots, u_1; r', r)$; admissible visits may thus be classified as undirected vertex sequences with unordered role pairs.* (b) *In a type-(i) admissible visit, both u_1 and u_k are terminal vertices, the roles r, r' are distinct, and the conditions read: first-edge color $= \overline{\mathrm{pc}(r)}$, last-edge color $= \overline{\mathrm{pc}(r')}$.*

Proof. (a) Reversal swaps first and last edges and the two endpoint constraints. (b) The roles occupy $u_1 \neq u_k$, hence are distinct; the rest is the definition together with the fact that with two colors "$\neq \mathrm{pc}(r)$" means "$= \overline{\mathrm{pc}(r)}$". □

6 Inflation and the inflation lemma

Definition 6.1 (Inflation)**.** Let $a \geq 2$, let τ be a permutation of $\{1, \ldots, a\}$, and let $\pi_1, \ldots, \pi_a$ be permutations of sizes $b_1, \ldots, b_a \geq 2$, $N = \sum_i b_i$. Put $s_i = \sum_{j<i} b_j$ (position offsets) and $t_i = \sum_{j:\ \tau(j)<\tau(i)} b_j$ (value offsets). The *inflation* $\sigma = \tau[\pi_1, \ldots, \pi_a] \in S_N$ is

$$\sigma(s_i + r) = t_i + \pi_i(r), \qquad i = 1, \ldots, a, \quad 0 \leq r < b_i.$$

The value intervals $[t_i, t_i + b_i - 1]$ partition $\{0, \ldots, N-1\}$, so σ is a bijection. Each position block $B_i = [s_i, s_i + b_i - 1]$ is an interval block of σ with value interval $[t_i, t_i + b_i - 1]$ and pattern π_i, so Lemma 5.2 applies to all blocks simultaneously. Edges of H_σ with endpoints in different blocks are *cross edges*.

Lemma 6.2. *$H_{\tau[\pi_1,\ldots,\pi_a]}$ has exactly $2(a-1)$ cross edges, counted with multiplicity: $a-1$ P-colored (one per consecutive pair of position blocks) and $a-1$ V-colored (one per consecutive pair of value intervals). Each block is incident to at most 4 cross edges, attaching at its terminals as in Lemma 5.2(b), and each terminal role of each block carries at most one cross edge.*

Proof. A P-edge $\{p, p+1\}$ is a cross edge iff $p+1 \in \{s_2, \ldots, s_a\}$: $a-1$ edges. A V-edge $\{\sigma^{-1}(v), \sigma^{-1}(v+1)\}$ is a cross edge iff $v, v+1$ lie in different value intervals, which happens for exactly the $a-1$ values v ending a value interval. The attachment statement is Lemma 5.2(b); per block, each of the four port positions is met by at most one cross edge. □

Lemma 6.3 (Visit decomposition)**.** *Let $\sigma = \tau[\pi_1, \ldots, \pi_a]$, $a \geq 2$, and let Q be a PC path in H_σ. Partition the vertex sequence of Q into maximal runs of consecutive vertices lying in a common block; call these runs* visits*. Then:*

(a) Consecutive visits are joined by exactly one cross edge; distinct consecutive pairs use distinct cross edges. Hence if Q has q visits, then $q - 1 \leq 2(a-1)$, i.e. $q \leq 2a - 1$.

(b) Every visit that is neither the first nor the last is an admissible visit of the copy of H_{π_i} on its block; in particular it has at most $M(\pi_i)$ vertices.

(c) The first and the last visit have at most $\max_i b_i$ *vertices each.*

Proof. (a) The edge of Q joining the last vertex of one visit to the first vertex of the next has endpoints in different blocks (maximality of runs), so it is a cross edge. Q has distinct vertices, hence pairwise distinct edges, so the $q-1$ connecting edges are distinct cross edges; there are $2(a-1)$ of them in all (Lemma 6.2).

(b) Let $W = u_1, \ldots, u_k$ be an interior visit in block B_i. There is an *entry* cross edge e_{in} of Q at u_1 and an *exit* cross edge e_{out} at u_k. By Lemma 6.2, e_{in} attaches at a terminal role r at u_1 with color $\mathrm{pc}(r)$, and e_{out} at a role r' at u_k with color $\mathrm{pc}(r')$. All edges of Q between vertices of W lie in B_i, so by Lemma 5.2(a) W is a PC path in H_{π_i}. If $k \geq 2$: the PC condition at u_1 (between e_{in} and the first edge of W) forces the first edge of W to have color $\neq \mathrm{pc}(r)$, similarly at u_k; so W with roles r, r' is admissible of type (i). If $k = 1$: the PC condition at u_1 between e_{in} and e_{out} forces $\mathrm{pc}(r) \neq \mathrm{pc}(r')$: admissible of type (ii).

(c) A visit lies in one block. □

Theorem 6.4 (Inflation Lemma)**.** *For* $a \geq 2$*,*

$$\rho\big(\tau[\pi_1, \ldots, \pi_a]\big) \;\leq\; (2a-3)\,\max_i M(\pi_i) \;+\; 2\,\max_i b_i.$$

Proof. Let Q attain ρ with q visits. If $q = 1$, then $\rho \leq \max_i b_i$. If $q \geq 2$, then by Lemma 6.3 the $q - 2 \leq 2a - 3$ interior visits have at most $\max_i M(\pi_i)$ vertices each and the two end visits at most $\max_i b_i$ each; visits partition the vertices of Q. □

Theorem 6.4 holds for *every* outer permutation τ. The problem is thereby reduced to finding patterns with small transit number relative to their size.

7 The gadget: classification of admissible visits of H_{T_b}

Definition 7.1. For even $b \geq 6$ let $T_b \in S_b$ be given by

$$T_b(0) = 0, \qquad T_b(1) = b-1, \qquad T_b(i) = i \;\; (2 \leq i \leq b-2), \qquad T_b(b-1) = 1.$$

Thus T_b^{-1} fixes 0 and each $v \in [2, b-2]$ and swaps $1 \leftrightarrow b-1$ (T_b is an involution). Recall $b \geq 6$.

Lemma 7.2 (Inventory)**.** *The edges of* H_{T_b} *are exactly:*

- doubled pairs $\{i, i+1\}$ *for* $2 \leq i \leq b-3$ *(each carrying one P- and one V-edge);*
- P-only edges $\{0, 1\}$*,* $\{1, 2\}$*,* $\{b-2, b-1\}$*;*
- V-only edges $\{0, b-1\}$*,* $\{2, b-1\}$*,* $\{1, b-2\}$*.*

The full edge lists at the relevant vertices are:

vertex $0:$	$\mathrm{P}\,\{0,1\}$; $\mathrm{V}\,\{0,b{-}1\}$ (*degree* 2),
vertex $1:$	$\mathrm{P}\,\{0,1\}$, $\mathrm{P}\,\{1,2\}$; $\mathrm{V}\,\{1,b{-}2\}$ (*unique V-edge*),
vertex $b{-}1:$	$\mathrm{P}\,\{b{-}2,b{-}1\}$; $\mathrm{V}\,\{0,b{-}1\}$, $\mathrm{V}\,\{2,b{-}1\}$ (*unique P-edge*),
vertex $2:$	$\mathrm{P}\,\{1,2\}$; *doubled* $\{2,3\}$; $\mathrm{V}\,\{2,b{-}1\}$,
vertex $b{-}2:$	*doubled* $\{b{-}3,b{-}2\}$; $\mathrm{P}\,\{b{-}2,b{-}1\}$; $\mathrm{V}\,\{1,b{-}2\}$,
vertex $i,\ 3 \leq i \leq b-3:$	*doubled* $\{i{-}1,i\}$ *and* $\{i,i{+}1\}$ *only.*

The terminals are $p_0 = v_0 =$ *vertex* 0*,* $v_1 =$ *vertex* 1*,* $p_1 =$ *vertex* $b-1$*.*

Proof. P-edges are $\{i, i+1\}$, $0 \le i \le b-2$. The V-edges, indexed by $v = 0, \dots, b-2$, are $\{T_b^{-1}(v), T_b^{-1}(v+1)\}$:

$$v=0:\ \{0, b{-}1\}; \qquad v=1:\ \{b{-}1, 2\}; \qquad 2 \le v \le b-3:\ \{v, v{+}1\}; \qquad v=b{-}2:\ \{b{-}2, 1\}.$$

A pair $\{i, i+1\}$ is doubled iff it appears in both lists: exactly for $2 \le i \le b-3$. For $b \ge 6$: $\{0,1\}$ doubled would need $b-1=1$; $\{1,2\}$ would need $b-3=1$; $\{b-2, b-1\}$ would need $b-3=1$; all false, and the three long V-edges join non-adjacent positions since $b-1, b-3 \ge 3$. The per-vertex lists and the terminals ($T_b^{-1}(0)=0$, $T_b^{-1}(b-1)=1$) follow by inspection. $\square$

Let $S = \{2, 3, \dots, b-2\}$ (the *segment*), $|S| = b-3$. By Lemma 7.2, the sub-multigraph on S is the doubled path $2 - 3 - \cdots - (b-2)$; the only edges from S to its complement are the two at vertex 2 (P $\{1,2\}$, V $\{2, b-1\}$) and the two at vertex $b-2$ (P $\{b-2, b-1\}$, V $\{1, b-2\}$); and no terminal lies in S.

Lemma 7.3 (Segment behavior)**.** *Let Q be a PC path in H_{T_b}.*

(a) While inside S, Q moves monotonically in position (each interior vertex of S is adjacent only to its two position-neighbors), and Q can leave S only from vertex 2 or vertex $b-2$.

(b) Suppose Q enters S at one end (2 or $b-2$) by an edge of color c, at a moment when no vertex of S has yet been visited, and later leaves S. Then either it leaves immediately from the entry vertex (using the other external edge there, of color $\bar{c}$), or it traverses all of S to the far end; in the latter case its $b-4$ internal edges alternate colors starting with $\bar{c}$, so (since b is even, making $b-4$ even) the last internal edge has color c, and any edge by which Q then leaves S has color $\bar{c}$.

Proof. (a) is immediate from the inventory: a simple path cannot reverse inside a path graph. For (b): leaving from the entry vertex means the run in S is that single vertex; the two external edges at 2 (resp. $b-2$) have different colors, and the PC condition forces the exit edge to differ from the entry edge, i.e. to be the other one, of color $\bar{c}$. Otherwise the run proceeds monotonically and exits only at the far end, having covered all of S with internal edges $g_1, \dots, g_{b-4}$. The PC condition at the entry vertex gives $g_1 = \bar{c}$ (both colors are available on every doubled pair; the PC condition forces the alternation), alternation gives $g_m = \bar{c}$ for odd m and $g_m = c$ for even m, and $b-4$ is even, so $g_{b-4} = c$; the PC condition at the far end forces the exit edge to have color $\neq c$. $\square$

7.1 The classification theorem

Theorem 7.4 (Classification of admissible visits)**.** *Let $b \ge 6$ be even. Up to reversal (Lemma 5.5), the graph H_{T_b} has exactly five admissible visits. They are the following, each with the role pair shown:*

	vertices	*edges (colors)*	*role pair*
(V1)	0	—	$\{p_0, v_0\}$
(V2)	$0,\ b-1$	$\mathrm{V}\{0, b{-}1\}$	$\{p_0, p_1\}$
(V3)	$0,\ 1$	$\mathrm{P}\{0, 1\}$	$\{v_0, v_1\}$
(V4)	$b-1,\ 0,\ 1$	$\mathrm{V}\{0, b{-}1\},\ \mathrm{P}\{0,1\}$	$\{p_1, v_1\}$
(V5)	$b-1,\ 2,\ 1$	$\mathrm{V}\{2, b{-}1\},\ \mathrm{P}\{1,2\}$	$\{p_1, v_1\}$

In particular: the role pairs $\{p_0, v_1\}$ and $\{v_0, p_1\}$ admit no visit; the only visit for $\{p_0, p_1\}$ is (V2); the only visit for $\{v_0, v_1\}$ is (V3); and $M(T_b) = 3$.

Proof. That each listed visit is admissible is a direct check from Lemma 7.2. (V1): vertex 0 carries p_0 and v_0 with different port colors. (V2): first = last edge $\mathrm{V} \neq \mathrm{pc}(p_0) = \mathrm{pc}(p_1) = \mathrm{P}$. (V3): first = last edge $\mathrm{P} \neq \mathrm{pc}(v_0) = \mathrm{pc}(v_1) = \mathrm{V}$. (V4), (V5): first edge $\mathrm{V} \neq \mathrm{pc}(p_1)$, last edge $\mathrm{P} \neq \mathrm{pc}(v_1)$, and the middle vertex alternates the colors. The role pairs are forced: e.g. in (V2) the first edge is V, so the role at 0 has port color $\neq$ V, i.e. p_0; the same one-line check fixes each role pair.

It remains to show no other admissible visit exists. Type-(ii) visits need a vertex carrying two roles of different port colors; by Lemma 7.2 the only such vertex is $0 = p_0 = v_0$, giving (V1). A type-(i) visit starts at a terminal vertex with a constrained first color (Lemma 5.5(b)). The terminal vertices are $0, 1, b-1$, giving exactly four (start vertex, first color) *stems*:

(A) $(0, \mathrm{V})$ [role p_0], (B) $(0, \mathrm{P})$ [role v_0], (C) $(1, \mathrm{P})$ [role v_1], (D) $(b-1, \mathrm{V})$ [role p_1].

Every admissible visit, read from its first vertex, extends one of these stems, so it suffices to enumerate, for each stem, every PC continuation, recording the moments at which the path stands at a terminal vertex with a last-edge color admissible for a role there. We use the *forcing facts* from Lemma 7.2: (F1) vertex 0 has exactly one P-edge ($\{0,1\}$) and one V-edge ($\{0, b-1\}$); (F2) vertex 1 has exactly one V-edge ($\{1, b-2\}$); (F3) vertex $b-1$ has exactly one P-edge ($\{b-2, b-1\}$); plus the segment behavior of Lemma 7.3. No vertex of S is a terminal, so a path stranded in S can never end an admissible visit there.

Stem (A): start 0, first edge V. By (F1) the first edge is $\mathrm{V}\{0, b-1\}$; the path stands at $b-1$, last color V. *Record:* role p_1 at $b-1$ has pc = P $\neq$ V, so $(0, b-1)$ is admissible: this is (V2). Continue: next $\neq$ V, so by (F3) $\mathrm{P}\{b-2, b-1\}$; enter S at $b-2$, entry color P, S untouched. By Lemma 7.3(b):

- *Immediate exit at $b-2$* by $\mathrm{V}\{1, b-2\}$: stand at 1, last color V; the only role at 1 is v_1 (pc = V): not admissible. Continue from 1: next $\neq$ V, so a P-edge; $\{0,1\}$ excluded (0 visited), leaving $\mathrm{P}\{1,2\}$ into S at 2. From 2, next $\neq$ P: $\mathrm{V}\{2, b-1\}$ excluded ($b-1$ visited), leaving the V-side of $\{2,3\}$; from 3 the path moves monotonically up and strands (exits are 2, behind it, and $b-2$, visited). No admissible visit.

- *Full crossing to* 2: arrival at 2 by color P (Lemma 7.3(b) with $c = \mathrm{P}$); exit must be V; the only V-external edge at 2 is $\{2, b-1\}$ with $b-1$ visited. Strands. No admissible visit.

Stem (A) yields exactly (V2).

Stem (B): start 0, first edge P. By (F1) the first edge is $\mathrm{P}\{0,1\}$; stand at 1, last color P. *Record:* role v_1 at 1 has pc = V $\neq$ P: admissible: (V3). Continue: next $\neq$ P, so by (F2) $\mathrm{V}\{1, b-2\}$; enter S at $b-2$, entry color V, S untouched. By Lemma 7.3(b):

- *Immediate exit at $b-2$* by $\mathrm{P}\{b-2, b-1\}$: stand at $b-1$, last color P; the only role at $b-1$ is p_1 (pc = P): not admissible. Continue from $b-1$: next $\neq$ P, so a V-edge; $\{0, b-1\}$ excluded (0 visited), leaving $\mathrm{V}\{2, b-1\}$ into S at 2. From 2, next $\neq$ V: $\mathrm{P}\{1,2\}$ excluded (1 visited), leaving the P-side of $\{2,3\}$; monotone up and strands ($b-2$ visited). No admissible visit.

- *Full crossing to* 2: arrival color V; exit must be P; only P-external edge at 2 is $\{1,2\}$, and 1 is visited. Strands. No admissible visit.

Stem (B) yields exactly (V3).

Stem (C): start 1**, first edge** P**.** The P-edges at 1 are $\{0,1\}$ and $\{1,2\}$; two branches.

- *Branch* $1 \to 0$. At 0, last color P. *Record:* v_0 (pc $=$ V) admissible: (V3) reversed. Continue: next $\neq$ P, by (F1) V$\{0, b-1\}$; at $b-1$, last color V. *Record:* p_1 admissible: (V4) reversed. Continue: next $\neq$ V, by (F3) P$\{b-2, b-1\}$ into S at $b-2$, entry color P, S untouched. Immediate exit by V$\{1, b-2\}$: 1 visited. Full crossing: arrival at 2 color P, exit must be V: only $\{2, b-1\}$, visited. Strands; branch closed.

- *Branch* $1 \to 2$. Enter S at 2 by P$\{1,2\}$, entry color P, S untouched. Immediate exit by V$\{2, b-1\}$: at $b-1$, last color V. *Record:* p_1 admissible: (V5) reversed. Continue from $b-1$: next $\neq$ V, by (F3) P$\{b-2, b-1\}$ into S at $b-2$ – but S is now touched (vertex 2): from $b-2$ the non-P continuations are V$\{1, b-2\}$ (1 visited) and the V-side of $\{b-3, b-2\}$, after which the path moves monotonically down and strands at 3 (2 visited). Branch closed. Alternatively, full crossing $2 \to b-2$ with entry color P arrives at $b-2$ by color P; exit must be V: only $\{1, b-2\}$, 1 visited. Strands; branch closed.

Stem (C) yields exactly (V3), (V4), (V5) (reversed).

Stem (D): start $b-1$**, first edge** V**.** The V-edges at $b-1$ are $\{0, b-1\}$ and $\{2, b-1\}$; two branches.

- *Branch* $b-1 \to 0$. At 0, last color V. *Record:* p_0 admissible: (V2) reversed. Continue: next $\neq$ V, by (F1) P$\{0,1\}$; at 1, last color P. *Record:* v_1 admissible: (V4). Continue: next $\neq$ P, by (F2) V$\{1, b-2\}$ into S at $b-2$, entry color V, S untouched. Immediate exit by P$\{b-2, b-1\}$: $b-1$ visited. Full crossing: arrival at 2 color V, exit must be P: only $\{1,2\}$, 1 visited. Strands; branch closed.

- *Branch* $b-1 \to 2$. Enter S at 2 by V$\{2, b-1\}$, entry color V, S untouched. Immediate exit by P$\{1,2\}$: at 1, last color P. *Record:* v_1 admissible: (V5). Continue from 1: next $\neq$ P, by (F2) V$\{1, b-2\}$ into S at $b-2$ – S touched (vertex 2): from $b-2$ the non-V continuations are P$\{b-2, b-1\}$ ($b-1$ visited) and the P-side of $\{b-3, b-2\}$, then monotone descent stranding at 3 (2 visited). Branch closed. Alternatively, full crossing $2 \to b-2$ with entry color V arrives by color V; exit must be P: only $\{b-2, b-1\}$, $b-1$ visited. Strands; branch closed.

Stem (D) yields exactly (V2), (V4), (V5).

Collecting the four stems: every admissible visit is one of (V1)–(V5) up to reversal, each realized only with the listed role pair (checked at each *Record* step). No admissible visit realizes $\{p_0, v_1\}$ or $\{v_0, p_1\}$, and the maximum size is 3, attained by (V4) and (V5). $\square$

Corollary 7.5. *$M(T_b) = 3$ for every even $b \geq 6$, and $c^* := \liminf_n \rho_{\min}(n)/n$ satisfies $c^* \leq 6/b$ for every even $b \geq 6$; hence $c^* = 0$ and the question of Section 1 has a negative answer.*

Proof. $M(T_b) = 3$ is part of Theorem 7.4. Taking $\pi_i = T_b$ for all i in Theorem 6.4 and letting $a \to \infty$: $\rho_{\min}(ab)/(ab) \leq \big((2a-3) \cdot 3 + 2b\big)/(ab) \to 6/b$. $\square$

Remark 7.6 (The parity is essential)**.** For *odd* b, $b-4$ is odd, so the last internal edge of a full crossing has color $\bar{c}$ and the exit edge may have color c. The route $0 \xrightarrow{\mathrm{P}} 1 \xrightarrow{\mathrm{V}} b-2 \rightsquigarrow 2 \xrightarrow{\mathrm{V}\{2,b-1\}} b-1$ (full crossing in the middle) then becomes properly colored, an admissible visit for $\{v_0, p_1\}$ of size b. Exact computation (Section 11) confirms $M(T_b) = b$ for odd $b \in \{7, 9, 11, 13\}$ and $M(T_b) = 3$ for even $6 \le b \le 100$, and confirms the *exact list* of Theorem 7.4 for even $6 \le b \le 14$ by independent brute-force enumeration (ancillary file `verify_M.py`).

Corollary 7.7 (Two visits to one T_b-block)**.** *Let $\sigma = \tau[\pi_1, \dots, \pi_a]$ be an inflation, B_i a block whose pattern π_i equals T_b for some even $b \ge 6$, and Q a PC path in H_σ. Then the interior visits of Q to B_i have total size at most* 4*.*

Proof. By Lemma 6.2, B_i has at most four incident cross edges, one per role; each interior visit uses two distinct ones (entry and exit), and every cross edge is used at most once. Hence there are at most two interior visits; if at most one, its size is at most $M(T_b) = 3 \le 4$. If two, they use all four cross edges, so their role pairs partition $\{p_0, p_1, v_0, v_1\}$, and they are vertex-disjoint (disjoint runs of the simple path Q). By Theorem 7.4 the three partitions give:

- $\{p_0, p_1\}/\{v_0, v_1\}$: the only visits are (V2)$= (0, b-1)$ and (V3)$= (0, 1)$, which share vertex 0 – impossible;
- $\{p_0, v_1\}/\{v_0, p_1\}$: no visits exist for either pair – impossible;
- $\{p_0, v_0\}/\{p_1, v_1\}$: the $\{p_0, v_0\}$ visit is (V1) at vertex 0, and the $\{p_1, v_1\}$ visit is (V4) or (V5); since (V4)$= (b-1, 0, 1)$ contains vertex 0 it cannot be disjoint from (V1), so the disjoint pair is (V1) with (V5), of sizes 1 and 3: total 4. (In every case the total is at most 4.)

□

8 Proof of the main theorem

We prove the bound advertised in Theorem 1.1. The family is $\sigma_{a,b} = \mathrm{id}_a[T_b, \dots, T_b]$, i.e. $\sigma_{a,b}(jb + r) = jb + T_b(r)$, and the proof combines the two-visit bound for one T_b-block (Corollary 7.7) with the inflation count of Section 6.

Theorem 8.1 (Main bound)**.** *For every even $b \ge 6$ and $a \ge 2$, $\rho(\sigma_{a,b}) \le 4a + 2b$. Consequently $\rho_{\min}(n) \le 8\sqrt{n} + 20$ for every $n \ge 100$, so $\rho_{\min}(n) = O(\sqrt{n})$ and $\rho_{\min}(n)/n \to 0$.*

Proof. The family bound. Let Q be a PC path in $H_{\sigma_{a,b}}$ with q visits. If $q = 1$ then $|Q| \le b$. Otherwise, by Lemma 6.3 the first and last visits have at most b vertices each, and by Corollary 7.7 the interior visits contribute at most 4 per block, i.e. at most $4a$ in total. Hence $\rho(\sigma_{a,b}) \le 4a + 2b$.

The bound on $\rho_{\min}(n)$. Let $b = 2\lfloor \sqrt{n}/2 \rfloor$, so b is even, $\sqrt{n} - 2 < b \le \sqrt{n}$, and $b \ge 10$ since $n \ge 100$. Let $a = \lfloor n/b \rfloor \ge 2$ and $s = n - (a-1)b$, so $b \le s < 2b$. Define $\sigma \in S_n$ as the inflation (identity outer permutation) of:

- if s even: $a-1$ copies of T_b and one copy of T_s (s even, $s \ge b \ge 10$);
- if s odd: $a-1$ copies of T_b, one copy of T_{s-3}, and one block of size 3 with identity pattern $\iota_3 = (0, 1, 2)$ ($s-3$ even, $s-3 \ge b-2 \ge 8$, since s odd and b even force $s \ge b+1$).

The number of blocks is at most $a+1 \le n/b+1$, and each block is one of T_b, T_s, T_{s-3}, or ι_3. Interior visits contribute at most 4 per T-block (Corollary 7.7, which applies verbatim to T_b, T_s, T_{s-3}) and at most $2 \cdot 3 = 6$ for the single ι_3-block (at most two interior visits, by Lemma 6.2, of size at most 3 each). The two end visits contribute at most $2(2b-1)$, since the largest block has size at most $2b-1$. Hence

$$\rho_{\min}(n) \;\le\; 4\Big(\frac{n}{b}+1\Big)+6+2(2b-1) \;=\; \frac{4n}{b}+4b+8 \;\le\; \big(4\sqrt{n}+10\big)+4\sqrt{n}+8 \;\le\; 8\sqrt{n}+20,$$

using $4n/(\sqrt{n}-2) = 4\sqrt{n}+8\sqrt{n}/(\sqrt{n}-2) \le 4\sqrt{n}+10$ for $\sqrt{n} \ge 10$. □

Remark 8.2 (A weaker bound without the full classification)**.** If one uses only the value $M(T_b) = 3$ and not the two-visit refinement, the Inflation Lemma (Theorem 6.4) already gives $\rho(\sigma_{a,b}) \le 6a+2b$ and, by the same choice of blocks (each of transit number at most 3), $\rho_{\min}(n) \le (2m-3)\cdot 3 + 2(2b-1) \le 6n/b + 4b - 5 \le 10\sqrt{n}+10$ for $n \ge 100$. The refinement to $8\sqrt{n}+20$ is what the two-visit corollary buys.

Proposition 8.3 (Explicit long path: the family is $\Theta(a+b)$)**.** *For every even $b \ge 6$ and $a \ge 2$, $\rho(\sigma_{a,b}) \ge 2a+2b-7$.*

Proof. Write (j,ℓ) for the vertex at position $jb+\ell$ ($0 \le j < a$, $0 \le \ell < b$), so block j is a copy of H_{T_b} on $\{(j,0),\dots,(j,b-1)\}$, with values $(j,0) \mapsto jb$, $(j,1) \mapsto jb+b-1$, $(j,\ell) \mapsto jb+\ell$ for $2 \le \ell \le b-2$, and $(j,b-1) \mapsto jb+1$. Besides the internal edges (Lemma 7.2 in each block), the cross edges are the P-edges $\{(j,b-1),(j+1,0)\}$ and the V-edges $\{(j,1),(j+1,0)\}$. Within block j note the V-edges $\{(j,0),(j,b-1)\}$ and $\{(j,2),(j,b-1)\}$.

The following walk is a PC path; colors are indicated over each step, and each edge exists by the inventory.

- *Stage 1 (block 0).* Start at $(0,b-2)$ and descend the segment to $(0,2)$ along the doubled run, choosing colors V, P, V, …; this uses $b-4$ edges, and since $b-4$ is even the last (into $(0,2)$) has color P. Then take V $\{(0,2),(0,b-1)\}$, then the P-cross edge to $(1,0)$.
- *Stage 2 (blocks $1,\dots,a-2$).* From $(j,0)$ (entered by a P-cross edge) take V $\{(j,0),(j,b-1)\}$, then the P-cross edge to $(j+1,0)$; repeat.
- *Stage 3 (block $a-1$).* From $(a-1,0)$ (entered by a P-cross edge) take V $\{(a-1,0),(a-1,b-1)\}$, then P $\{(a-1,b-2),(a-1,b-1)\}$, then descend the segment from $(a-1,b-2)$ to $(a-1,2)$ choosing colors V, P, …, stopping at $(a-1,2)$.

All vertices are distinct (stages use disjoint blocks; within a stage the listed vertices are distinct), and consecutive edge colors differ at every junction. The vertex count is $(b-2)$ in stage 1 ($b-3$ segment vertices plus $(0,b-1)$), $2(a-2)$ in stage 2, and $(b-1)$ in stage 3 ($(a-1,0)$, $(a-1,b-1)$, and the $b-3$ segment vertices), totalling $2a+2b-7$. The path is checked mechanically, edge by edge, in the ancillary file `verify_path.py`. □

Remark 8.4 (Constants; exact values)**.** Together, Theorem 8.1 and Proposition 8.3 give $2(a+b)-7 \le \rho(\sigma_{a,b}) \le 4a+2b$: the family is $\Theta(a+b)$, and with $a \asymp b$ realizes $\Theta(\sqrt{n})$, so the upper bound of Theorem 1.1 is achieved by this family up to the constant. Exact machine computation (Section 11) gives $\rho(\sigma_{a,b}) = 2a+2b-4$ at every tested point up to $n = 720$; we conjecture this holds for all $a \ge 3$, even $b \ge 6$, but we neither prove nor use it. Optimizing the proven upper bound $4a+2b$ at $n = ab$ with $b \asymp \sqrt{2n}$ gives $\rho_{\min}(n) \le 4\sqrt{2n}+O(1) \approx 5.66\sqrt{n}$ along such n.

9 The transit floor: no pattern has transit number below three

The main theorem rests on patterns whose transit number is 3 while their size grows. Within the inflation calculus, one parameter is decisive: how small the transit number of a pattern can be made. Define

$$\mu(b) \;:=\; \min_{\pi \in S_b} M(\pi) \qquad (b \geq 2).$$

Any family of patterns with μ-optimal transit numbers gives the best interior-visit bookkeeping the Inflation Lemma can produce. This section determines μ completely.

Theorem 9.1 (Transit floor)**.** *$\mu(b) = 3$ for every $b \geq 3$. (For $b = 2$, both permutations have $M = 2$, so $\mu(2) = 2$.)*

The upper bound is a construction, the lower bound a structure theorem; we prove them in turn. Together they say the gadgets of this paper are optimal in the transit parameter: no choice of inflation blocks can beat $M = 3$ at any size.

9.1 The upper bound: gadgets at every size

Proposition 9.2. *$\mu(b) \leq 3$ for every $b \geq 3$.*

Proof. For $b = 3$ the inequality is trivial: an admissible visit has at most b vertices, so $M(\pi) \leq 3$ for every $\pi \in S_3$. For even $b \geq 6$, Corollary 7.5 gives $M(T_b) = 3$. For odd $b \geq 7$, let

$$T'_b \;=\; (0,\; 1,\; b{-}1,\; 3,\; 4,\; \ldots,\; b{-}2,\; 2) \in S_b$$

be the identity with the values 2 and $b{-}1$ exchanged – a sibling of T_b, which exchanges 1 and $b{-}1$. Appendix B classifies the admissible visits of $H_{T'_b}$ completely, in the manner of Theorem 7.4: up to reversal there are exactly five, of sizes $1, 3, 3, 3, 3$, so $M(T'_b) = 3$. (The parity mechanism is the same doubled-run obstruction as in Lemma 7.3, with the run shortened by one vertex, which is exactly what reverses the good parity from even b to odd b.) The two remaining sizes $b = 4, 5$ are settled by the exhaustive enumeration already reported in Section 11: $\min_{\pi \in S_b} M(\pi) = 3$ for $4 \leq b \leq 9$; for $b = 5$ the pattern $T'_5 = (0, 1, 4, 3, 2)$ attains it, with admissible-visit list (0), $(0, 1, 2)$, $(0, 1, 4)$, $(2, 1, 4)$, $(2, 3, 4)$. □

9.2 The lower bound: every pattern admits a visit of size three

The lower bound cannot be proved by a local or greedy argument: for the identity permutation the *smallest* witness visit produced below is a terminal-to-terminal path through all b vertices, so no argument that exhibits a bounded-size certificate can succeed uniformly. The proof instead builds the witness globally, from one matching inside each color class. Three elementary lemmas about matchings in paths do all the work. Throughout, $\pi \in S_b$ with $b \geq 3$; the *P-path* is the Hamiltonian path $0 - 1 - \cdots - (b-1)$ of P-edges and the *V-path* is the Hamiltonian path $\pi^{-1}(0) - \pi^{-1}(1) - \cdots - \pi^{-1}(b-1)$ of V-edges. For a matching M contained in one of these paths, we say M *exposes* the vertices it does not cover.

Lemma 9.3 (Matching unions)**.** *Let M_P be a matching contained in the P-edges and M_V a matching contained in the V-edges, and let U be their union, keeping colors (if a doubled pair contributes an edge to each, U retains both parallel edges). Then:*

(a) *every vertex has degree* ≤ 2 *in* U*; more precisely, the degree of* x *in* U *equals the number of the two matchings that cover* x*;*

(b) *the connected components of* U *are simple paths (possibly single vertices) and cycles, and along any component the edges alternate between* M_P *and* M_V*, hence alternate colors: every component is properly colored;*

(c) *the degree-*1 *vertices are exactly those covered by exactly one of the two matchings, and the unique* U*-edge at such a vertex belongs to the matching covering it. If* U *has exactly* 2ℓ *vertices of degree* 1*, then* U *has exactly* ℓ *components that are paths with at least one edge, their ends pairing up the degree-*1 *vertices.*

Proof. (a) Each matching contributes at most one edge per vertex. Consecutive edges of a walk in U share a vertex, so they cannot lie in the same matching; hence they alternate between M_P and M_V, and since M_P-edges are P-colored and M_V-edges V-colored, the colors alternate, giving (b) (maximum degree 2 makes every component a path or a cycle; a doubled pair with both copies in U is a 2-cycle component). The ends of the path components are the vertices of degree ≤ 1, giving (c). ☐

Lemma 9.4 (Prescribed exposure)**.** *Let* $Q = x_0 - x_1 - \cdots - x_{m-1}$ *be a path and* $S \subseteq \{x_0, \dots, x_{m-1}\}$*. A matching of* Q *exposing exactly* S *exists if and only if every maximal run of consecutive non-exposed vertices (in path order) has even length. In particular:*

(a) m *odd: a matching exposing exactly* $\{x_j\}$ *exists iff* j *is even;*

(b) m *even: a perfect matching exists, and a matching exposing exactly* $\{x_i, x_j\}$ $(i < j)$ *exists iff* i *is even and* j *is odd.*

Proof. If every run of non-exposed vertices is even, pair each run off consecutively. Conversely, a matching of a path covering a contiguous run of vertices bounded by exposed vertices must pair within the run, forcing even length. For (a), the runs have lengths j and $m - 1 - j$, both even iff j is even and m is odd; for (b), the runs have lengths i, $j - i - 1$, $m - 1 - j$, all even iff i is even, j is odd, m is even. ☐

Lemma 9.5 (Exposure trick)**.** *Suppose* M_P *exposes exactly the vertex set* S_P *within the P-path and* M_V *exposes exactly* S_V *within the V-path, and let* U *be their union. Then:*

(a) *the degree-*1 *vertices of* U *are exactly* $(S_P \cup S_V) \setminus (S_P \cap S_V)$*; the vertices of* $S_P \cap S_V$ *are isolated; all others have degree* 2*;*

(b) (opposite exposure $\Rightarrow$ odd size ≥ 3) *if a path component of* U *has one end* $x \in S_P \setminus S_V$ *and the other end* $y \in S_V \setminus S_P$*, then its end edges are V-colored at* x *and P-colored at* y*, its edge count is even, and it has at least* 2 *edges: its size is odd and* ≥ 3*;*

(c) (same exposure $\Rightarrow$ even size ≥ 2) *if both ends lie in* $S_P \setminus S_V$*, the end edges are V-colored at both ends, the edge count is odd, and the size is even and* ≥ 2*, with size exactly* 2 *precisely when the two ends are joined by a single* M_V*-edge (symmetrically with the colors exchanged for two ends in* $S_V \setminus S_P$*).*

Proof. (a) restates Lemma 9.3(a). For (b): $x \in S_P \setminus S_V$ is covered only by M_V, so its unique U-edge is V-colored; likewise the unique edge at y is P-colored (Lemma 9.3(c)). Alternation with first color V and last color P forces an even number of edges, and the component has at least one edge since its ends have degree 1. If it had exactly one edge $e = \{x, y\}$, then e, being x's edge, lies in M_V; but then M_V covers y, contradicting $y \in S_V$. So the edge count is even and ≥ 2, and the size (edges plus one) is odd and ≥ 3. For (c): both end edges are V-colored, so alternation $\mathrm{V} \cdots \mathrm{V}$ gives an odd edge count ≥ 1; the size is even and ≥ 2, and size 2 is exactly the single-edge case, which forces that edge into M_V. □

Theorem 9.6 (Floor lower bound)**.** *For every $\pi \in S_b$ with $b \geq 3$, the graph H_π has a type-(i) admissible visit of size ≥ 3. Hence $M(\pi) \geq 3$.*

Proof. Write $v_0 = \pi^{-1}(0)$ and $v_1 = \pi^{-1}(b-1)$; these are distinct vertices since $b \geq 2$. All matchings below live inside the P-path or the V-path, and their existence is by Lemma 9.4, using the following indices: vertex 0 has position-index 0 and vertex $b-1$ has position-index $b-1$ in the P-path; v_0 has value-index 0 and v_1 has value-index $b-1$ in the V-path.

Case 1: b odd. Both P-path indices $0, b-1$ and both V-path indices $0, b-1$ are even. Set $c := v_1$ if $v_1 \neq 0$, and otherwise $c := v_0$; in the latter case $\pi(0) = b-1 \neq 0$, so $v_0 = \pi^{-1}(0) \neq 0$. Either way $c \neq 0$, and c has V-path index 0 or $b-1$, both even. By Lemma 9.4(a) choose M_P exposing exactly $\{0\}$ (concretely $\{1, 2\}, \{3, 4\}, \ldots, \{b-2, b-1\}$) and M_V exposing exactly $\{c\}$. Then $S_P = \{0\}$ and $S_V = \{c\}$ are disjoint, so by Lemma 9.5(a) the union U has exactly two degree-1 vertices, 0 and c, hence exactly one path component, from 0 to c. By Lemma 9.5(b) it is a properly colored path of odd size ≥ 3 whose first edge is V-colored at vertex 0 and whose last edge is P-colored at c. Assign the role p_0 at vertex 0 (port color P $\neq$ V) and the role v_1 or v_0 at c (port color V $\neq$ P): a type-(i) admissible visit of size ≥ 3.

Case 2: b even and $\{\pi(0), \pi(b-1)\} \neq \{2k, 2k+1\}$ for every k. By Lemma 9.4(b) choose M_V to be the perfect matching of the V-path (value pairs $(0, 1), (2, 3), \ldots, (b-2, b-1)$) and M_P exposing exactly $\{0, b-1\}$ (indices 0 even and $b-1$ odd). Then $S_P = \{0, b-1\}$, $S_V = \emptyset$: the union has exactly one path component, from 0 to $b-1$, with V-colored end edges at both ends, of even size ≥ 2 (Lemma 9.5(c)). Size 2 would mean the single edge $\{0, b-1\} \in M_V$, i.e. $\{\pi(0), \pi(b-1)\}$ is one of the value pairs $\{2k, 2k+1\}$ – excluded by hypothesis. So the size is even and ≥ 4; with roles p_0, p_1 (port colors P, end edges V) this is a type-(i) admissible visit of size ≥ 4.

Case 3: b even and $\{\pi(0), \pi(b-1)\} = \{2k, 2k+1\}$ for some k. By Lemma 9.4(b) choose M_P exposing exactly $\{0, b-1\}$ and M_V exposing exactly $\{v_0, v_1\}$ (value-indices 0 even and $b-1$ odd; the exposed value pairs are $(1, 2), (3, 4), \ldots, (b-3, b-2)$). Consider the coincidence set $T := \{0, b-1\} \cap \{v_0, v_1\}$.

Case 3a: $T = \emptyset$. There are four degree-1 vertices $0, b-1, v_0, v_1$, hence exactly two path components pairing them up. If some component joins a P-terminal to a V-terminal, Lemma 9.5(b) makes it an admissible visit of odd size ≥ 3 (roles with opposite port colors at its ends) — done. Otherwise one component joins 0 to $b-1$ and the other joins v_0 to v_1, both of even size ≥ 2 (Lemma 9.5(c)). If the $0 \rightsquigarrow b-1$ component had size 2, then $\{0, b-1\} \in M_V$, so $\{\pi(0), \pi(b-1)\}$ would be one of the value pairs $\{2j+1, 2j+2\}$ – an odd-aligned consecutive pair – while the case hypothesis makes it even-aligned; a two-element set of consecutive integers cannot be both. So that component has even size ≥ 4, and with roles p_0, p_1 it is an admissible visit of size ≥ 4 – done.

Case 3b: $|T| = 1$. Let t be the coincident vertex ($t \in S_P \cap S_V$ is isolated in U), x the other P-terminal and y the other V-terminal; $x \neq y$ (else $|T| = 2$). The degree-1 vertices are exactly x and y, so U has one path component, from x to y, which by Lemma 9.5(b) is an admissible visit of odd size ≥ 3 (a P-role at x with V-colored end edge, a V-role at y with P-colored end edge) – done.

Case 3c: $|T| = 2$. Then $\{v_0, v_1\} = \{0, b-1\}$, i.e. $\{\pi(0), \pi(b-1)\} = \{0, b-1\}$, which is a set of consecutive integers only if $b = 2$ – contradicting $b \geq 4$ in the presence of the case-3 hypothesis. So this sub-case is vacuous.

Cases 1–3 cover all $\pi \in S_b$, $b \geq 3$. □

Remark 9.7 (The witness is necessarily global)**.** For $\pi = \mathrm{id}_b$ with b odd, Case 1 produces the full alternating path $0 - 1 - \cdots - (b-1)$: the witness has size b, not size 3. This is forced, not an artifact: in H_{id_b} every admissible visit joining terminals at the two ends must traverse the whole doubled path. A proof of Theorem 9.6 that exhibits a bounded-radius certificate around a terminal therefore cannot exist, which is why the matching-union construction, whose witness is a *component of a globally defined graph*, is the right tool.

Proof of Theorem 9.1. Combine Proposition 9.2 and Theorem 9.6; for $b = 2$, enumerate directly (each of the two patterns admits a size-2 visit between its two P-roles or V-roles, and size 3 is impossible on 2 vertices). □

Remark 9.8 (Consequences for the method)**.** Theorem 9.1 answers the calibration question raised in Section 12: transit numbers cannot be pushed below 3, so within the inflation framework the per-visit cost of this paper's gadgets is optimal, and improvements to Theorem 1.1 must come from the visit *count* or from outside the inflation method. The construction of Theorem 9.6 is validated mechanically, independently of its proof, in the ancillary artifact (Section 11).

Remark 9.9 (The spectrum of transit numbers; data)**.** Exhaustive computation of $\{M(\pi) : \pi \in S_b\}$ for $b \leq 8$ (Section 11) shows that the attained set is $\{3, 4, \ldots, b\}$ for even b but $\{3\} \cup \{5, \ldots, b\}$ for odd b: the value 4 is not attained at odd $b \leq 8$. The maximum $M(\pi) = b$ is always attained (e.g. by the identity). We record the gap at 4 as data; we have no proof that it persists for odd $b > 8$.

10 Block counts do not bound the longest properly colored path

Recall from Definition 4.1 the ± 1-block count $b(\sigma)$ and the two elementary facts that frame the lower-bound question: $\rho \geq s_{\max}$ always, and $s_{\max} \cdot b(\sigma) \geq n$. Any inequality of the form $\rho(\sigma) \geq c\, b(\sigma)$ with an absolute $c > 0$ would therefore give $\rho_{\min}(n) \geq c\sqrt{n}$ and settle Conjecture 12.1. The inequality with $c = 1$ holds exhaustively for $n \leq 12$ (Section 11). This section shows that the route is closed completely: first we locate the exact threshold where $\rho \geq b(\sigma)$ fails – it is $n = 14$, far below the $n = 240$ counterexamples visible inside the family $\sigma_{a,b}$ – and then we prove that *no* positive constant works: $\inf_\sigma \rho(\sigma)/b(\sigma) = 0$. The witnesses are *block-free* permutations – $b(\sigma) = n$, every block a singleton – with $\rho = \Theta(\sqrt{n})$, so the failure is as extreme as it could be.

10.1 The exact threshold for $\rho \geq b(\sigma)$

Theorem 10.1. *Let $\sigma^\dagger = (2, 0, 3, 1, 7, 5, 9, 4, 8, 6, 12, 10, 13, 11) \in S_{14}$. Then $\sigma^\dagger$ is block-free, $b(\sigma^\dagger) = 14$, and $\rho(\sigma^\dagger) = 13$, so $\rho \geq b(\sigma)$ fails at $n = 14$. Moreover, exhaustive computation (Section 11) shows:*

(a) $\rho(\sigma) \geq b(\sigma)$ for every $\sigma \in S_n$ with $n \leq 12$;

(b) every one of the 831,283,558 block-free permutations in S_{13} is properly-colored Hamiltonian ($\rho = 13$);

(c) at $n = 14$, exactly 16 block-free permutations are not properly-colored Hamiltonian; each of them violates $\rho \geq b(\sigma)$.

Hence $n = 14$ is the exact threshold within the block-free class, and no counterexample of any kind exists for $n \leq 12$. (Whether a counterexample with $b(\sigma) < n$ exists at $n = 13$ we have not determined.)

Proof. Block-freeness of $\sigma^\dagger$ is a direct check: the consecutive value differences are

$$-2,\ 3,\ -2,\ 6,\ -2,\ 4,\ -5,\ 4,\ -2,\ 6,\ -2,\ 3,\ -2,$$

none equal to ± 1, so every ± 1-block is a singleton and $b(\sigma^\dagger) = 14$. The values $\rho(\sigma^\dagger) = 13$ and the census counts in (a)–(c) are exhaustive computations, reproducible from the ancillary artifact; the single-instance value $\rho(\sigma^\dagger) = 13$ is confirmed by the reference solvers of Section 11 (a PC path on 13 vertices exists and the exact search over all states returns no larger one). $\square$

Remark 10.2. Theorem 10.1(b) and (c) also settle the extreme case $b(\sigma) = n$ left open in earlier discussions of this problem: block-free permutations need *not* be properly-colored Hamiltonian, but the first failures occur only at $n = 14$. The block-reversal data of Section 4 (the half-bound, true through $n = 12$, false at $n = 18$) and Theorem 10.1 (the block bound, true through $n \leq 12$, false at $n = 14$) are two instances of the same lesson: in this problem family, exhaustive verification at small n is weak evidence.

10.2 Block-free chains with $\rho = \Theta(\sqrt{n})$

The threshold witnesses have deficit 1. To make the failure asymptotic we need block-free permutations whose ρ grows much more slowly than n; we build them as inflations whose blocks are copies of a *modular staircase* pattern.

Definition 10.3 (Staircase pattern)**.** Call s *admissible* if $s \geq 9$ is odd and $s \not\equiv 1 \pmod 3$ (equivalently $s \equiv 3, 5 \pmod 6$). For admissible s put $m = s - 1$ (so m is even and $3 \nmid m$) and define $\theta_s \in S_s$ by

$$\theta_s(0) = m, \qquad \theta_s(i) = 3i \bmod m \ \ (1 \leq i \leq m-1), \qquad \theta_s(m) = 0.$$

For $k \geq 1$ let $\Theta_{s,k} := \mathrm{id}_k[\theta_s, \ldots, \theta_s] \in S_{sk}$ be the k-fold chain (Definition 6.1 with identity outer permutation). We write *piece j* ($1 \leq j \leq k$) for the j-th position block, *vertex i of piece j* for position $(j-1)s + i$, and *junction j* for the boundary between pieces j and $j+1$.

For example $\theta_9 = (8, 3, 6, 1, 4, 7, 2, 5, 0)$ and $\theta_{11} = (10, 3, 6, 9, 2, 5, 8, 1, 4, 7, 0)$.

Lemma 10.4 (Staircase basics)**.** *Let s be admissible and $k \geq 1$. Then:*

(a) θ_s is a permutation of $\{0, \ldots, m\}$;

(b) θ_s is block-free, and so is every chain: $b(\Theta_{s,k}) = sk$;

(c) the cross edges of $\Theta_{s,k}$ at junction j $(1 \leq j \leq k-1)$ are exactly: the P-edge joining vertex m of piece j to vertex 0 of piece $j+1$, and the V-edge joining vertex 0 of piece j to vertex m of piece $j+1$. Consequently every cross edge incident to piece j attaches at vertex 0 or vertex m of the piece: at vertex 0, the P-edge of junction $j-1$ and the V-edge of junction j; at vertex m, the V-edge of junction $j-1$ and the P-edge of junction j (omitting nonexistent junctions).

Proof. (a) Multiplication by 3 is a bijection on $\mathbb{Z}_m$ since $3 \nmid m$, and it maps $\{1, \ldots, m-1\}$ onto itself (only $0 \mapsto 0$); together with $\theta_s(0) = m$ and $\theta_s(m) = 0$ all values $0, \ldots, m$ occur exactly once.

(b) Consecutive value differences of θ_s: for $1 \leq i \leq m-2$, $\theta_s(i+1) - \theta_s(i) \equiv 3 \pmod{m}$, so the difference is 3 or $3-m$; at the ends, $\theta_s(1) - \theta_s(0) = 3-m$ and $\theta_s(m) - \theta_s(m-1) = -(m-3)$ (since $3(m-1) \bmod m = m-3$). As $m \geq 8$, no difference is ± 1, so every ± 1-block of θ_s is a singleton. In the chain, differences within a piece equal differences of θ_s (the values shift by a constant per piece), and across junction j the difference is $\big(js+m\big) - (j-1)s = s+m \geq 2$ – the last position of piece j holds the piece's bottom value $(j-1)s$ and the first position of piece $j+1$ holds that piece's top value $js+m$. So every block of $\Theta_{s,k}$ is a singleton.

(c) By Lemma 6.2 the cross edges are one P-edge and one V-edge per junction. The P-edge joins the last position of piece j (vertex m) to the first position of piece $j+1$ (vertex 0); its value gap is $s+m \geq 2$, so it is P-only. The V-edge joins $\Theta^{-1}(js-1)$ to $\Theta^{-1}(js)$, i.e. the position of piece j's top value to the position of piece $j+1$'s bottom value; within a piece the top value sits at vertex 0 ($\theta_s(0) = m$) and the bottom value at vertex m ($\theta_s(m) = 0$), so the V-edge joins vertex 0 of piece j to vertex m of piece $j+1$, a non-adjacent position pair: V-only. The attachment statement reads off these two descriptions at junctions $j-1$ and j. $\square$

Lemma 10.5 (Parity)**.** *H_{θ_s} is bipartite with parts the even and the odd positions: every edge joins an even position to an odd one. Consequently every PC path in H_{θ_s} between vertices 0 and m has an even, positive number of edges, and hence its first and last edges have* different *colors.*

Proof. P-edges join consecutive positions. For the V-edges, let $g := 3^{-1} \bmod m$; then g is odd, since m is even and an even g would make $3g \bmod m$ even, while $3g \equiv 1$. The V-edge of value v joins $\theta_s^{-1}(v)$ to $\theta_s^{-1}(v+1)$. For $1 \leq v \leq m-2$ these are $gv \bmod m$ and $g(v+1) \bmod m$, whose difference is $\equiv g \pmod{m}$; since m is even, adding an odd residue modulo m flips the parity of a position. For $v = 0$ the edge is $\{m, g\}$ and for $v = m-1$ it is $\{m-g, 0\}$; in both, one entry is even (m, 0) and the other odd (g, $m-g$). So every edge flips parity. Vertices 0 and m are both even, so any path between them has an even number of edges, and it is positive since $0 \neq m$. A properly colored path alternates colors, so with an even edge count its first and last edges differ in color. $\square$

Theorem 10.6 (Chain collapse)**.** *For every admissible s and every $k \geq 4$,*

$$\rho(\Theta_{s,k}) \;\leq\; 4s + 2(k-4), \qquad \textit{while} \qquad b(\Theta_{s,k}) = sk.$$

Proof. Let Q be a PC path in $H_{\Theta_{s,k}}$, decomposed into visits as in Lemma 6.3. Each visit lies in one piece; classify the visits: an *end visit* contains an endpoint of Q (it has at most one of entry/exit cross edge); every other visit has both an entry and an exit cross edge, which attach to its piece and hence belong to the piece's two junctions (Lemma 10.4(c)) – call the visit a *return* if they belong to the same junction and a *through visit* otherwise.

Through visits have size 1. Let W be a through visit of piece j, with entry and exit cross edges at junctions $j-1$ and j in some order. By Lemma 5.2(a) the piece induces a color-isomorphic copy of H_{θ_s}, and by Lemma 10.4(c) the four cross edges attach at the piece's vertices 0 and m with the colors listed there. Four cases. If the entry is the junction-$(j-1)$ P-edge (at vertex 0) and the exit is the junction-j P-edge (at vertex m), then W is a PC path from 0 to m; if it has ≥ 2 vertices, the PC condition at its ends forces its first and its last internal edge to be V-colored – equal colors – contradicting Lemma 10.5; and it cannot have one vertex since $0 \neq m$. The V-in/V-out case is symmetric (both end edges P-colored). If the entry is the junction-$(j-1)$ P-edge (at vertex 0) and the exit is the junction-j V-edge (also at vertex 0), then the first and last vertex of W coincide, so W is the single vertex 0; the PC condition there holds as the two cross edges have different colors. Symmetrically, V-in at m with P-out at m gives the single vertex m. So every through visit that exists at all has exactly one vertex.

Returns occur only at the extremities. Suppose a visit R of piece j enters and exits through the two cross edges of junction $j-1$ (a *left return*). Those are the only two edges crossing junction $j-1$, and a path uses each edge at most once, so Q decomposes as $A \cdot e_{\text{in}} \cdot R \cdot e_{\text{out}} \cdot B$ where A and B never cross junction $j-1$ and end (resp. begin) in piece $j-1$. Hence A and B lie entirely in pieces $< j$, and all vertices of Q in pieces $\geq j$ belong to R, which lies in piece j: piece j is the last piece Q touches. Two left returns would both have to sit at that last piece and jointly use four edges of one junction, which has only two; so Q has at most one left return, and symmetrically at most one *right return*, whose piece is the first piece Q touches.

Charging. The pieces touched by Q form a contiguous range (crossing between pieces uses a cross edge of the junction between them). Call a touched piece *special* if it contains an endpoint of Q or hosts a return: there are at most $2+1+1=4$ special pieces. A special piece contributes at most s vertices (its size). Every visit to a non-special touched piece is a through visit of size 1, and such a piece has at most 4 incident cross edges, each usable once, with each through visit consuming two: at most 2 visits, hence at most 2 vertices. With at most k touched pieces in all,

$$\rho(Q) \;\leq\; 4s + 2(k-4) \qquad (k \geq 4),$$

where the case of fewer than four special or touched pieces only improves the count. The block count is Lemma 10.4(b). $\square$

Remark 10.7 (Exactness; data)**.** Exact computation (Section 11) gives $\rho(\Theta_{9,k}) = 9k$ for $k \leq 4$ (the chains are properly-colored Hamiltonian up to four pieces: the transient is real) and, by certified-optimal constraint-solver runs, $\rho(\Theta_{9,k}) = 38, 40, 42$ at $k = 5, 6, 7$ ($n = 45, 54, 63$) and $\rho(\Theta_{15,4}) = 60$, $\rho(\Theta_{15,5}) = 62$: every certified value equals $4s + 2(k-4)$, so the bound of Theorem 10.6 is attained at every point checked. We use only the inequality.

Theorem 10.8 (No block floor)**.** $\inf_{\sigma} \dfrac{\rho(\sigma)}{b(\sigma)} = 0$*. In particular, for every constant* $c > 0$ *the inequality* $\rho(\sigma) \geq c\, b(\sigma)$ *fails, and for every constant* C *the inequality* $\rho(\sigma) \geq b(\sigma) - C$ *fails. Moreover there are block-free permutations* ($b(\sigma) = n$) *with* $\rho(\sigma) = O(\sqrt{n}\,)$*.*

Proof. Take $k = s$ in Theorem 10.6: the chain $\Theta_{s,s}$ on $n = s^2$ vertices is block-free with $\rho/b \le (6s-8)/s^2 \to 0$ as admissible $s \to \infty$, proving the infimum statement and the failure of $\rho \ge c\,b$ for every c. The deficit $b - \rho \ge s^2 - 6s + 8 \to \infty$ kills $\rho \ge b - C$ for every C. For the last statement, $\rho(\Theta_{s,s}) \le 6s - 8 = 6\sqrt{n} - 8$ with $b(\Theta_{s,s}) = n$. $\square$

Corollary 10.9 (Upper bound along the chains)**.** *For every admissible s, taking $k = 2s$ and $N = sk = 2s^2$,*

$$\rho_{\min}(N) \;\le\; \rho(\Theta_{s,2s}) \;\le\; 4s + 2(2s-4) \;=\; 4\sqrt{2}\,\sqrt{N} - 8,$$

so $\rho_{\min}(N) \le 4\sqrt{2}\,\sqrt{N} - 8$ on the infinite set $\{2s^2 : s \text{ admissible}\}$.

Remark 10.10 (Two extremes with one growth rate)**.** The leading constant $4\sqrt{2} \approx 5.66$ coincides with what the optimized gadget family already gives along its own special values of n (Remark 8.4). What is new is where the family sits: $\sigma_{a,b}$ has $b(\sigma) \approx 4\sqrt{n}$ blocks and $\rho/b \to \frac{1}{2}$, while the staircase chains sit at the opposite extreme $b(\sigma) = n$ with $\rho/b \to 0$. Permutations at *both* ends of the block-count spectrum thus collapse ρ to $\Theta(\sqrt{n})$ – which is why no statistic of the block structure alone can force long properly colored paths, and why Conjecture 12.1, if true, is best possible: no lower bound of order $\omega(\sqrt{n})$ can hold.

11 Computational verification

The asymptotic theorems and their constants are proved without computation; the computer-assisted claims are the exact values in the two counterexamples — $\rho(\sigma^*) = 10$ at $n = 18$ (Theorem 4.6) and $\rho(\sigma^\dagger) = 13$ at $n = 14$ (Theorem 10.1) – together with the exhaustive census counts of Theorem 10.1(a)–(c), the small-size base cases $b \in \{4, 5\}$ of Proposition 9.2, and the spectrum data of Remark 9.9. This section specifies the computations that independently verify the constructions and supply the exact small-case values, and records the empirical sharpness of the families. **All code, command lines, and output logs are provided as ancillary files with this submission** (directory `code/` in the repository, shipped as the standard ancillary directory `anc/` on arXiv; `code/README.md` maps each claim below to a script and a log, and `code/MANIFEST.sha256` lists SHA-256 checksums).

Algorithm. The basic object is an exact longest-PC-path search. Its state space is the set of triples (*visited set*, *current vertex*, *last edge color*); the initial states are all $(\{x, w\}, w, c)$ over edges $\{x, w\}$ of color c; a transition extends a state by an unvisited neighbor along an edge of the other color. The legal future extensions of a partial PC path depend only on its state, so distinct PC paths reaching the same state may be identified without loss; $\rho(\sigma)$ is then the maximum cardinality of the visited set over all *reachable* states, computed by an exhaustive search that visits every reachable state once (a memo set of states; no pruning). The method is exact by construction; its cost is the number of reachable states. That number is exponential in general – on unstructured instances we used the method only for $n \le 19$ – but is small on the structured families here, because the gadget admits relatively few reachable states: for $\sigma_{a,b}$ we measure 57,066 reachable states at $n = 80$, 5,012,186 at $n = 600$, and 7,273,586 at $n = 720$ (the logs report the count for every table row). The constrained variants computing $M(\pi)$ (fixed start/end terminals and first/last colors) are the same search with the initial and accepting states restricted.

Implementations. Three independent implementations were used: a plain recursive enumerator and a memoized iterative one (both Python, in the artifact), which agree on 300 random permutations of sizes 4–8; and a separately written C++ solver (also in the artifact) from a parallel verification effort, which agrees with the Python solvers on every instance checked by both, including exact values of $\rho(\sigma_{a,b})$ at $(a,b) \in \{(6,8),(5,10),(4,12)\}$ and the trap-family values of Appendix A at six points. In addition, the artifact contains a *minimal independent verifier* (`verify_M.py`, about 100 lines, written directly from Definition 5.4 with no shared code), which re-derives $M(T_b)$ and the complete list of admissible visits by brute force for $b \le 14$.

Verified statements. Each item states the claim, its range, and its log.

1. **Anchors of Section 4** (`validate2.log`, `search_small.log`): $\rho(\sigma_m) = 2m + 3$ for the block-reversal family at $m = 1,2,3,4$ ($n = 7,11,15,19$); $\rho(\sigma^*) = 10$ for the $n = 18$ counterexample; $\rho_{\min}(7) = 5$ and $\rho_{\min}(8) = 6$ by full enumeration of S_7, S_8.

2. **Classification** (`verify_M.log`): for even $b \in \{6,8,10,12,14\}$ the admissible visits of H_{T_b} are *exactly* (V1)–(V5) of Theorem 7.4; for odd $b \in \{7,9,11\}$, $M(T_b) = b$.

3. **The family at scale** (`check_all_even.log`): $M(T_b) = 3$ for *every* even $6 \le b \le 100$; $M(T_b) = b$ for odd $b \in \{7,9,11,13\}$ (`validate2.log`).

4. **Optimality of the gadget at small sizes** (`search_small.log`): full enumeration of S_b for $4 \le b \le 9$ gives $\min_{\pi \in S_b} M(\pi) = 3$ in every case; no gadget of size ≤ 9 beats transit number 3.

5. **Explicit lower-bound path** (`verify_path.log`): the path of Proposition 8.3 is checked edge-by-edge at seven (a,b) points including $(20,20)$.

6. **Exact values** (`endtoend.log`, `bigtable.log`): for $\sigma_{a,b}$:

a	b	n	ρ (exact)	$2a{+}2b{-}4$	$4a{+}2b$	ρ/n
10	8	80	32	32	56	0.400
12	8	96	36	36	64	0.375
12	12	144	44	44	72	0.306
16	12	192	52	52	88	0.271
20	12	240	60	60	104	0.250
16	16	256	60	60	96	0.234
24	16	384	76	76	128	0.198
20	20	400	76	76	120	0.190
30	16	480	88	88	152	0.183
24	20	480	84	84	136	0.175
30	20	600	96	96	160	0.160
36	20	720	108	108	184	0.150

Every exact value equals $2a + 2b - 4$ and sits inside the proven window $[2a{+}2b{-}7,\ 4a{+}2b]$ of (1); the two rows at $n = 480$ (differing in ρ) illustrate that $\rho(\sigma_{a,b})$ tracks a and b separately, not just their product. At $n = 720$ the ratio 0.150 is far below any ratio that the block-reversal experiments of Section 4 suggested was attainable.

7. **Exhaustive small-n statistics** (`logs/exh7.txt`–`exh12.txt`, C++ solver): the joint distribution of $(b(\sigma), \rho(\sigma))$ over all of S_n for $7 \leq n \leq 12$, used in Theorem 10.1(a) and Section 12; at $n = 12$ this enumerates all 479,001,600 permutations.

8. **Landscape** (exploratory; `probe.py`): under adjacent transpositions, $\sigma_{4,6}$ ($n = 24$, $\rho = 16$) is a strict local minimum of ρ, while greedy descent from random starts stalls around $\rho \approx 23$ – extremal permutations of this type are found by design, not by local search.

9. **The transit-floor construction** (`verify_floor.log`): the matching-union witness of Theorem 9.6 is built by an implementation of the exact case tree of the proof and validated against Definition 5.4 from scratch (proper coloring with respect to the actual edge sets, simplicity, endpoint roles, complementary end colors, size ≥ 3): exhaustively for *every* $\pi \in S_b$, $3 \leq b \leq 9$, and on random samples up to $b = 50$, with 0 failures.

10. **The odd gadget and the spectrum** (`verify_oddb.log`, `verify_spectrum.log`): the inventory, highway and crossing claims of Appendix B confirmed uniformly for odd $b \in \{7, 9, \ldots, 21\}$, and $M(T_b') = 3$ there and at the degenerate $b = 5$; as positive control the same checker reproduces $M(T_b) = 3$ (even b) and $M(T_b) = b$ (odd b) for the even-gadget family. Separately, exhaustive computation of $M(\pi)$ for every $\pi \in S_b$, $2 \leq b \leq 8$, yields $\mu(2) = 2$, $\mu(b) = 3$ for $3 \leq b \leq 8$, and the attained spectra of Remark 9.9 (`verify_spectrum.log`).

11. **The $n = 14$ threshold** (`verify_n14.log`): $\sigma^\dagger$ is block-free with $\rho(\sigma^\dagger) = 13$ (exact search, both reference solvers); the block-free censuses behind Theorem 10.1(b)–(c) – all 831,283,558 block-free permutations of S_{13} properly-colored Hamiltonian, exactly 16 failures at $n = 14$ – are reproducible from the census programs (`cpp/bfcensus.cpp` with log `blockfree13.txt`, hours of CPU time, a contracted matching-union Hamiltonicity criterion rather than the path solver, independently re-run once; and `n14_doublebad.py` for the exact count).

12. **Staircase chains** (`verify_stair.log`): block-freeness and the junction structure of Lemma 10.4 checked mechanically for admissible $s \leq 33$ and $k \leq 6$; exact $\rho(\Theta_{9,k}) = 9k$ for $k \leq 3$ by the reference solvers; the certified-optimal values of Remark 10.7 ($\rho = 36, 38, 40, 42$ at $n = 36, 45, 54, 63$; $60, 62$ for $s = 15$) obtained by an independent constraint-programming model (CP-SAT, `cpsat_rho.py` with log `cpsat_stair.txt`) that proves optimality, a solver unrelated to the state-space search.

Items 1–7 and 9–12 are exhaustive or certified within their stated ranges and reproducible by running the listed scripts; item 8 is exploratory.

12 Lower bounds and open problems

We work with ± 1-blocks (Definition 4.1); $b(\sigma)$ is their number and $s_{\max}(\sigma)$ the largest size. Two elementary facts frame the lower-bound question. First, the ± 1-blocks partition the positions, so

$$s_{\max}(\sigma) \cdot b(\sigma) \;\geq\; n, \qquad \text{hence} \qquad \max\bigl(s_{\max}(\sigma),\, b(\sigma)\bigr) \;\geq\; \sqrt{n}.$$

Second, $\rho(\sigma) \geq s_{\max}(\sigma)$ always (Proposition 4.2). Consequently an absolute $c > 0$ with $\rho(\sigma) \geq c\, b(\sigma)$ for all σ would give $\rho_{\min}(n) \geq c\sqrt{n}$ and, with Theorem 8.1, settle:

Conjecture 12.1. $\rho_{\min}(n) = \Theta(\sqrt{n})$.

Section 10 rules this out: no such c exists – indeed $\inf \rho/b = 0$, attained in the limit by block-free permutations whose ρ is itself $\Theta(\sqrt{n})$ (Theorem 10.8), with the exact finite threshold for $c = 1$ at $n = 14$ (Theorem 10.1). Two things survive on the positive side. First, the conjecture itself: both known collapsing families – the gadget inflations with $b \approx 4\sqrt{n}$ and the staircase chains with $b = n$ – bottom out at $\Theta(\sqrt{n})$, never below, and Corollary 10.9 pins the target from above at $4\sqrt{2}\sqrt{N} - 8$ infinitely often. Second, the unconditional floor $\rho \geq s_{\max}$, which degenerates for permutations with all blocks of bounded size – exactly the regime of Section 10's witnesses. Beyond it our methods produce no lower bounds at all. This is the state of the problem we most want to advertise:

Main open problem. Prove or disprove: $\rho_{\min}(n) \to \infty$.

The difficulty is easy to underestimate. The graphs H_σ are spanning, connected, of maximum degree four, and every vertex lies on two Hamiltonian monochromatic paths; the natural intuition is that so much connectivity must force alternating paths of unbounded length. But the quantities that make that intuition precise keep failing: the half-bound died at $n = 18$, the block bound at $n = 14$, the constant-factor block bound died outright (Theorem 10.8), and the surviving floor $\rho \geq s_{\max}$ is constant exactly on the candidate extremal families. A proof of $\rho_{\min}(n) \to \infty$ must therefore see something in H_σ that none of the block statistics see, and by Theorem 10.1 it cannot lean on small-n exhaustive evidence either. A disproof – a family with ρ bounded – would be at least as interesting, though we do not expect one (Conjecture 12.1).

On the gadget side, the calibration question is settled by Theorem 9.1: transit numbers cannot be made smaller than 3 at any size, so inflation with better blocks cannot improve the method's constants, and (heuristically) inflation alone cannot beat $\sqrt{n}$: the end-visit term ties the bound to the block size, and nesting inflations doubles the visit-count bound per level. What remains open there is the fine structure of the transit spectrum (Remark 9.9): is the value 4 unattained at every odd b?

Two further directions. First, the analogous question for the union of two Hamiltonian *cycles*: we expect the method to adapt – the port calculus acquires two wraparound cross edges and the visit count rises accordingly – but we have not carried out the analysis and do not claim the result. Second, the Lovász-side question (Section 1.3): for the graphs $G(n, \sigma)$ built from $\sigma_{a,b}$ or from the chains of Appendix A, is the maximum number of matching edges on a path still $\Theta(n)$? The properly colored relaxation is now known to be lossy; quantifying the loss on these explicit instances may indicate whether $f(n)$ is linear after all. (An explicit snake path through $G(ab, \sigma_{a,b})$ using $\geq a(b-3) = n - 3a$ matching edges, checked edge-by-edge at six points up to $n = 600$ in `verify_f_linear.py`, already shows the loss is large on this family: the matching-edge content stays linear while ρ drops to $\sqrt{n}$.)

A A second construction: trap chains

This appendix gives an independent proof of $\rho_{\min}(n) = O(\sqrt{n})$ by a different family, found in a parallel verification effort. It does not use Sections 5–7; conversely, the main text does not use this appendix. The mechanism is again a parity obstruction, but localized differently: here *large reversed blocks of odd size, flanked by singletons*, are traps that no properly colored path can cross.

Definition A.1. Let $C = (c_0, c_1, \ldots, c_r)$ be a composition of n with $c_t \geq 1$ and *no two consecutive entries equal to* 1. Put $P_t = c_0 + \cdots + c_{t-1}$, and let block t occupy positions and values $[P_t, P_t + c_t - 1]$, reversed:

$$\sigma_C(P_t + i) \;=\; P_t + c_t - 1 - i, \qquad 0 \leq i < c_t.$$

Write $L_t = P_t$ and $R_t = P_t + c_t - 1$ (the *ports* of block t).

Lemma A.2. *H_{σ_C} consists exactly of:*

(a) doubled pairs $\{p, p+1\}$ for all consecutive positions inside each block;

(b) for each $0 \leq t < r$, one P-only cross edge $\{R_t, L_{t+1}\}$;

(c) for each $0 \leq t < r$, one V-only cross edge $\{L_t, R_{t+1}\}$;

(d) nothing else.

*In particular the cross edges incident to block t attach as follows (*port–color crossover*): at L_t, a P-edge toward block $t-1$ and a V-edge toward block $t+1$; at R_t, a V-edge toward block $t-1$ and a P-edge toward block $t+1$ (omitting nonexistent neighbors). Non-port vertices carry no cross edges.*

Proof. (a) Inside a block, consecutive positions hold consecutive (descending) values: both a P-edge and a V-edge. (b) Positions R_t, L_{t+1} are consecutive; their values are P_t (block minimum) and $P_{t+1} + c_{t+1} - 1$ (block maximum), which differ by $c_t + c_{t+1} - 1 \geq 2$ since not both c_t, c_{t+1} equal 1: P-only. (c) The values $P_t + c_t - 1$ (maximum of block t, held at position L_t since the block descends) and $P_t + c_t = P_{t+1}$ (minimum of block $t+1$, held at R_{t+1}) are consecutive; the positions L_t, R_{t+1} differ by $c_t + c_{t+1} - 1 \geq 2$: V-only. (d) Positions in non-consecutive blocks are never adjacent; values in non-consecutive blocks never differ by 1 (value ranges are contiguous in block order); within consecutive blocks, (b)/(c) list the only position-adjacent (resp. value-adjacent) pairs not internal to a block. The crossover statement reads off (b) and (c) at blocks $t-1, t, t+1$. □

Lemma A.3. *Let Q be a PC path in H_{σ_C} and let t be a block containing no endpoint of Q. Then every maximal run of consecutive Q-vertices inside block t is either (i) a single port vertex, or (ii) a* full traversal *of all c_t vertices from one port to the other. In case (ii), if the entering cross edge has color χ, the exiting cross edge has color $\neq \chi$ if and only if c_t is odd.*

Proof. A run with no endpoint of Q has an entering and an exiting cross edge, which by Lemma A.2 attach only at ports; the run's internal edges lie in the block, whose internal graph is a doubled path. If the run enters and leaves at the same port it is that single vertex (re-reaching the entry port would repeat it); the two cross edges at one port have different colors (Lemma A.2), as the PC condition requires. Otherwise it goes from one port to the other through the entire internal path: a full traversal with $c_t - 1$ internal edges. The PC condition forces the internal colors to alternate starting with $\bar{\chi}$, so the last internal edge has color χ iff $c_t - 1$ is even; the exit edge differs from it, so it has color $\bar{\chi}$ iff c_t is odd. □

Lemma A.4 (Trap Lemma)**.** *Let block t of σ_C have odd size $c_t \geq 3$ with singleton neighbors $c_{t-1} = c_{t+1} = 1$ ($1 \leq t \leq r-1$). Let Q be a PC path with no endpoint in block t. Then Q uses at most 2 vertices of block t (at most one visit per port, and no full traversal).*

Proof. Write A for block t and s^-, s^+ for the single vertices of blocks $t-1$, $t+1$. By Lemma A.2 the four cross edges at A are

$$\text{at } L_A: \ \{s^-, L_A\} \text{ (P)}, \quad \{L_A, s^+\} \text{ (V)}; \qquad \text{at } R_A: \ \{s^-, R_A\} \text{ (V)}, \quad \{R_A, s^+\} \text{ (P)}.$$

Suppose Q makes a full traversal of A, entering by a cross edge of color χ; since c_t is odd it exits by a cross edge of color $\bar{\chi}$ (Lemma A.3). Four cases:

1. enter at L_A by $\{s^-, L_A\}$ ($\chi = \text{P}$): the exit is the V-colored cross edge at R_A, $\{s^-, R_A\}$ – but s^- was visited immediately before entering, a repetition;
2. enter at L_A by $\{L_A, s^+\}$ ($\chi = \text{V}$): the exit is $\{R_A, s^+\}$ – repeats s^+;
3. enter at R_A by $\{s^-, R_A\}$ ($\chi = \text{V}$): the exit is $\{s^-, L_A\}$ – repeats s^-;
4. enter at R_A by $\{R_A, s^+\}$ ($\chi = \text{P}$): the exit is $\{L_A, s^+\}$ – repeats s^+.

Each case contradicts simplicity of Q. So every visit is a single port vertex; there are two ports. □

Theorem A.5. *Let $L \geq 3$ be odd, $k \geq 1$, $0 \leq p \leq L+1$, and $C = (2, (L,1)^k, 3+p, 2)$, a composition of $n = (L+1)k + p + 7$ with no two consecutive 1's. Then*

$$\rho(\sigma_C) \ \leq \ 3k + 3L + p + 1.$$

Consequently $\rho_{\min}(n) \leq 7\sqrt{n} + O(1)$ for all sufficiently large n, by choosing L the largest odd integer $\leq \sqrt{n}$ and $k = \lfloor (n-7)/(L+1) \rfloor$, $p = n - 7 - (L+1)k$.

Proof. The blocks, in order, are: a 2-block; $A_1, s_1, A_2, s_2, \ldots, A_k, s_k$ where each A_t has size L and each s_t is a singleton; a $(3+p)$-block; a 2-block. For $2 \leq t \leq k$ the block A_t is flanked by the singletons s_{t-1} and s_t, so Lemma A.4 applies to it; A_1 is flanked on the left by the initial 2-block and is not trapped.

Fix a PC path Q and charge each block with the number of Q-vertices it contains; $\rho(Q)$ is the sum of the charges, bounded by category:

- each singleton s_t: charge ≤ 1; total $\leq k$;
- each trapped block A_t ($2 \leq t \leq k$) with no endpoint of Q: charge ≤ 2 (Lemma A.4); at most 2 trapped blocks may contain an endpoint of Q, each then with charge $\leq L$ (its size). Hence the trapped blocks contribute at most $2(k-1) + 2(L-2)$ in total (each endpoint block exceeds the trap bound 2 by at most $L-2$);
- A_1: charge $\leq L$;
- the three remaining blocks: charges ≤ 2, $\leq 3+p$, ≤ 2; total $\leq 7+p$.

Summing: $\rho(Q) \leq k + 2(k-1) + 2(L-2) + L + (7+p) = 3k + 3L + p + 1$. (If an endpoint lies in a non-trapped block, the corresponding excess term is not needed; the bound only improves.)

For the corollary: with $L \leq \sqrt{n}$ odd and maximal, $k \leq n/(L+1) \leq n/L$ and $p \leq L+1$, so $\rho \leq 3n/L + 4L + 4 = 7\sqrt{n} + O(1)$ using $L = \sqrt{n} - O(1)$. □

Remark A.6. The parity hypothesis is again essential: for even L, Lemma A.3 makes a full traversal *preserve* the boundary color, the four cases of Lemma A.4 become consistent, and computation shows the analogous even-L chains are fully traversable (ρ grows with slope 1). Exact computation (log `check_trap.log`, plus the parallel C++ solver) gives $\rho(\sigma_C) = 2k + 2(L-1)$ for $C = (2, (L, 1)^k, 3, 2)$, odd $L \in \{5, 7, 9, 13, 15, 21\}$ and all $k \geq 2$ within reach ($n \leq 57$): the same per-period slope 2 as the family of Section 8. We do not know whether the two families are related beyond sharing this slope.

B The odd gadget: classification of admissible visits of $H_{T'_b}$

This appendix proves the odd-size half of Proposition 9.2: for every odd $b \geq 7$, the permutation

$$T'_b \;=\; (0,\ 1,\ b{-}1,\ 3,\ 4,\ \ldots,\ b{-}2,\ 2) \in S_b \qquad \big(T'_b(2) = b{-}1,\ T'_b(b{-}1) = 2,\ T'_b(i) = i \text{ otherwise}\big)$$

has $M(T'_b) = 3$, by a complete classification of its admissible visits in the manner of Theorem 7.4. T'_b is an involution; $(T'_b)^{-1}$ fixes 0, 1 and each $i \in [3, b-2]$ and swaps $2 \leftrightarrow b-1$. The terminals (Definition 5.3) are

$$p_0 = v_0 = \text{vertex } 0, \qquad v_1 = (T'_b)^{-1}(b-1) = \text{vertex } 2, \qquad p_1 = \text{vertex } b-1.$$

Note the contrast with T_b: there the swap $1 \leftrightarrow b-1$ makes vertex 1 the terminal v_1; here the swap $2 \leftrightarrow b-1$ moves v_1 to vertex 2, one step further from the end of the position path, and this relabeling is the entire source of the parity reversal that makes the construction work at odd b.

Lemma B.1 (Inventory of $H_{T'_b}$)**.** *For odd $b \geq 7$, the edges of $H_{T'_b}$ are exactly:*

- doubled pairs $\{0,1\}$ *and* $\{k, k+1\}$ *for* $3 \leq k \leq b-3$*;*
- P-only edges $\{1,2\}$*,* $\{2,3\}$*,* $\{b-2, b-1\}$*;*
- V-only edges $\{1, b-1\}$*,* $\{2, b-2\}$*,* $\{3, b-1\}$.

The full edge lists at the special vertices are:

$$\begin{array}{ll}
\textit{vertex } 0: & \textit{doubled } \{0,1\} \textit{ only (a leaf: its unique neighbor is vertex } 1\textit{),}\\
\textit{vertex } 1: & \mathrm{P}\,\{0,1\},\ \mathrm{P}\,\{1,2\};\quad \mathrm{V}\,\{0,1\},\ \mathrm{V}\,\{1,b{-}1\},\\
\textit{vertex } 2: & \mathrm{P}\,\{1,2\},\ \mathrm{P}\,\{2,3\};\quad \mathrm{V}\,\{2,b{-}2\}\quad (\textit{unique V-edge}),\\
\textit{vertex } 3: & \mathrm{P}\,\{2,3\},\ \textit{doubled}\ \{3,4\};\quad \mathrm{V}\,\{3,b{-}1\},\\
\textit{vertex } b{-}2: & \textit{doubled}\ \{b{-}3,b{-}2\};\ \mathrm{P}\,\{b{-}2,b{-}1\};\quad \mathrm{V}\,\{2,b{-}2\},\\
\textit{vertex } b{-}1: & \mathrm{P}\,\{b{-}2,b{-}1\};\quad \mathrm{V}\,\{1,b{-}1\},\ \mathrm{V}\,\{3,b{-}1\}\quad (\textit{unique P-edge}),\\
\textit{vertex } i,\ 4 \leq i \leq b-3: & \textit{doubled}\ \{i{-}1,i\} \textit{ and } \{i,i{+}1\} \textit{ only.}
\end{array}$$

Proof. P-edges are $\{i, i+1\}$, $0 \leq i \leq b-2$. The V-edges, indexed by $v = 0, \ldots, b-2$, are $\{(T'_b)^{-1}(v), (T'_b)^{-1}(v+1)\}$:

$$v = 0:\ \{0,1\}; \qquad v = 1:\ \{1, b{-}1\}; \qquad v = 2:\ \{b{-}1, 3\};$$
$$3 \leq v \leq b-3:\ \{v, v{+}1\}; \qquad v = b-2:\ \{b{-}2, 2\}.$$

The consecutive-position pairs appearing in both lists are exactly $\{0, 1\}$ and $\{k, k+1\}$ for $3 \le k \le b-3$: doubled. The remaining V-edges $\{1, b-1\}, \{3, b-1\}, \{2, b-2\}$ join non-adjacent positions for $b \ge 7$: V-only. The remaining P-edges $\{1, 2\}, \{2, 3\}, \{b-2, b-1\}$ are P-only. The per-vertex lists follow by inspection. $\square$

We use three forcing facts, the analogues of (F1)–(F3) in the proof of Theorem 7.4: **(F0)** vertex 0 is a leaf whose unique neighbor is vertex 1, reached by the doubled pair $\{0, 1\}$ – so in any simple PC path, vertex 0 is an endpoint; **(F1)** vertex $b-1$ has a unique P-edge, $\{b-2, b-1\}$; **(F2)** vertex 2 has a unique V-edge, $\{2, b-2\}$.

Let $D = \{3, 4, \ldots, b-2\}$ (the *highway*), $|D| = b-4$. By Lemma B.1, the sub-multigraph on D is the doubled path $3 - 4 - \cdots - (b-2)$; the only edges from D to its complement are the two at vertex 3 (P $\{2, 3\}$, V $\{3, b-1\}$) and the two at vertex $b-2$ (P $\{b-2, b-1\}$, V $\{2, b-2\}$); and no terminal lies in D.

Lemma B.2 (Highway behavior)**.** *Let Q be a PC path in $H_{T'_b}$, $b \ge 7$ odd.*

- *(a) While inside D, Q moves monotonically in position, and Q can leave D only from vertex 3 or vertex $b-2$.*
- *(b) Suppose Q enters D at one end (3 or $b-2$) by an edge of color c, at a moment when no vertex of D has yet been visited, and later leaves D. Then either it leaves immediately from the entry vertex by the other external edge there, of color $\bar{c}$, or it traverses all of D to the far end; in the latter case its $b-5$ internal edges alternate colors starting with $\bar{c}$, and since b is* odd*, $b-5$ is* even*, so the last internal edge has color c and any edge by which Q then leaves D has color $\bar{c}$.*

Proof. Identical in form to Lemma 7.3, applied to D: (a) is immediate since a simple path cannot reverse inside a doubled path graph and the boundary vertices of D are 3 and $b-2$; the two external edges at each of 3 and $b-2$ have different colors, so an immediate exit uses the other one, of color $\bar{c}$. A full crossing has $b-5$ internal edges (a path on $b-4$ vertices); the PC condition at the entry gives the first internal edge color $\bar{c}$, alternation gives the last internal edge color c (as $b-5$ is even), and the PC condition at the far end forces any exit edge to color $\bar{c}$. $\square$

Note the parity contrast with T_b: the segment of Lemma 7.3 has $b-4$ internal edges (even iff b even), while the highway here has $b-5$ (even iff b odd), because the relabeling detached vertex 2 from the doubled run. The parity that serves even b in Theorem 7.4 is exactly the parity that serves odd b here.

Theorem B.3 (Classification, odd gadget)**.** *Let $b \ge 7$ be odd. Up to reversal, the admissible visits of $H_{T'_b}$ are exactly the following, each with the role pair shown:*

	vertices	*edges (colors)*	*role pair*
(W1)	0	—	$\{p_0, v_0\}$
(W2)	$0,\ 1,\ 2$	$\mathrm{V}\{0,1\},\ \mathrm{P}\{1,2\}$	$\{p_0, v_1\}$
(W3)	$0,\ 1,\ b-1$	$\mathrm{P}\{0,1\},\ \mathrm{V}\{1,b{-}1\}$	$\{v_0, p_1\}$
(W4)	$2,\ 1,\ b-1$	$\mathrm{P}\{1,2\},\ \mathrm{V}\{1,b{-}1\}$	$\{v_1, p_1\}$
(W5)	$2,\ 3,\ b-1$	$\mathrm{P}\{2,3\},\ \mathrm{V}\{3,b{-}1\}$	$\{v_1, p_1\}$

In particular every admissible visit has at most 3 *vertices, and* $M(T_b') = 3$*, attained by (W2)–(W5).*

Proof. Each listed visit is admissible. (W1): vertex 0 carries p_0 and v_0 with different port colors – type (ii). (W2): the visit $(0,1,2)$ takes the V-copy of the doubled pair $\{0,1\}$, then P$\{1,2\}$; first edge V $\neq \mathrm{pc}(p_0) =$ P, last edge P $\neq \mathrm{pc}(v_1) =$ V. (W3): P-copy of $\{0,1\}$, then V$\{1,b-1\}$; first P $\neq \mathrm{pc}(v_0)$, last V $\neq \mathrm{pc}(p_1)$. (W4): P$\{1,2\}$ then V$\{1,b-1\}$; first P $\neq \mathrm{pc}(v_1)$, last V $\neq \mathrm{pc}(p_1)$. (W5): P$\{2,3\}$ then V$\{3,b-1\}$; the same check. The role pairs are forced by the first/last edge colors as in Theorem 7.4. (Note vertex 1 is *not* a terminal here – it is an interior vertex of (W2)–(W4).)

No others exist. A type-(ii) visit needs a vertex carrying two roles of different port colors; the only vertex carrying two roles is $0 = p_0 = v_0$: (W1). A type-(i) visit starts at a terminal vertex with a constrained first color (Lemma 5.5(b)); the terminal vertices are $0, 2, b-1$, and the possible (start, first color) stems are

$$\text{(A) } (0,\mathrm{V})\ [p_0], \quad \text{(B) } (0,\mathrm{P})\ [v_0], \quad \text{(C) } (2,\mathrm{P})\ [v_1], \quad \text{(D) } (b-1,\mathrm{V})\ [p_1]$$

(vertex 2 carries only v_1, which requires first color P; vertex $b-1$ carries only p_1, requiring first color V). We enumerate the PC continuations of each stem, recording every moment the path stands at a terminal vertex with an admissible last-edge color; since no vertex of D is a terminal, a path stranded inside D ends no admissible visit.

Stem (A): start 0**, first edge** V**.** By (F0) the first edge is the V-copy of $\{0,1\}$; at 1, last color V; no role at 1. Continue with a P-edge: $\{0,1\}$ is excluded (0 visited), so P$\{1,2\}$; at 2, last color P. *Record:* v_1 at 2 has pc $=$ V $\neq$ P: this is (W2). Continue: next $\neq$ P, so by (F2) V$\{2,b-2\}$, entering D at $b-2$ with color V, D untouched. By Lemma B.2(b): *immediate exit* by P$\{b-2,b-1\}$: at $b-1$, last color P; the only role there is p_1 with pc $=$ P: not admissible. Continue: next $\neq$ P, a V-edge at $b-1$: $\{1,b-1\}$ has 1 visited, so V$\{3,b-1\}$ into D at 3; from 3 the non-V continuations are P$\{2,3\}$ (2 visited) or the P-side of $\{3,4\}$, after which the path ascends monotonically and strands ($b-2$ visited). *Full crossing* to 3: arrival color V, so the exit must be P; the only external P-edge at 3 is $\{2,3\}$, with 2 visited. Strands. *Stem (A) yields exactly (W2).*

Stem (B): start 0**, first edge** P**.** By (F0) the first edge is the P-copy of $\{0,1\}$; at 1, last P; no role. Continue with a V-edge: $\{0,1\}$ excluded, so V$\{1,b-1\}$; at $b-1$, last V. *Record:* p_1 admissible: (W3). Continue: next $\neq$ V, by (F1) P$\{b-2,b-1\}$ into D at $b-2$, color P, D untouched. *Immediate exit* by V$\{2,b-2\}$: at 2, last V; role v_1 has pc $=$ V: not admissible. Continue: next $\neq$ V, a P-edge $\{1,2\}$ (1 visited) or P$\{2,3\}$ into D at 3; from 3, next $\neq$ P is V$\{3,b-1\}$ ($b-1$ visited) or the V-side of $\{3,4\}$, then monotone ascent strands ($b-2$ visited). *Full crossing* to 3: arrival P, exit must be V; only $\{3,b-1\}$, with $b-1$ visited. Strands. *Stem (B) yields exactly (W3).*

Stem (C): start 2**, first edge** P**.** The P-edges at 2 are $\{1,2\}$ and $\{2,3\}$. *Branch* $2 \to 1$*:* at 1, last P; no role. Continue with a V-edge: either V$\{0,1\}$ – at 0, last V; *record:* p_0 (pc $=$ P) admissible: (W2) reversed; by (F0) the only continuation returns to 1, visited: dead end – or V$\{1,b-1\}$ – at $b-1$, last V; *record:* p_1 admissible: (W4). Continue: next $\neq$ V, by (F1) P$\{b-2,b-1\}$ into D at $b-2$, color P, D untouched; immediate exit by V$\{2,b-2\}$ has 2 visited; full crossing to 3 arrives with P, exit must be V, only $\{3,b-1\}$ with $b-1$ visited. Strands. *Branch* $2 \to 3$*:* enters D at 3, color P, D untouched. Immediate exit by V$\{3,b-1\}$: at $b-1$, last V; *record:* p_1 admissible: (W5). Continue: next $\neq$ V, by (F1) P$\{b-2,b-1\}$ into D at $b-2$

– but D is now touched (vertex 3): the non-P continuations at $b-2$ are V$\{2, b-2\}$ (2 visited) and the V-side of $\{b-3, b-2\}$, then monotone descent strands at 4 (3 visited). Alternatively, full crossing $3 \to b-2$ arrives with color P; the exit must be V, and the only external V-edge at $b-2$ is $\{2, b-2\}$, with 2 visited. Strands. *Stem (C) yields exactly (W2) reversed, (W4), (W5).*

Stem (D): start $b-1$, first edge V**.** The V-edges at $b-1$ are $\{1, b-1\}$ and $\{3, b-1\}$. *Branch $b-1 \to 1$:* at 1, last V; no role. Continue with a P-edge: either P$\{0,1\}$ – at 0, last P; *record:* v_0 (pc = V) admissible: (W3) reversed; the only continuation returns to 1: dead end – or P$\{1,2\}$ – at 2, last P; *record:* v_1 admissible: (W4) reversed. Continue: next $\neq$ P, by (F2) V$\{2, b-2\}$ into D at $b-2$, color V, D untouched; immediate exit by P$\{b-2, b-1\}$ has $b-1$ visited; full crossing to 3 arrives with V, exit must be P, only $\{2,3\}$ with 2 visited. Strands. *Branch $b-1 \to 3$:* enters D at 3, color V, D untouched. Immediate exit by P$\{2,3\}$: at 2, last P; *record:* v_1 admissible: (W5) reversed. Continue: next $\neq$ P, by (F2) V$\{2, b-2\}$ into D at $b-2$ – D touched (vertex 3): the non-V continuations at $b-2$ are P$\{b-2, b-1\}$ ($b-1$ visited) and the P-side of $\{b-3, b-2\}$, then monotone descent strands at 4. Alternatively, full crossing $3 \to b-2$ arrives with color V; the exit must be P, only $\{b-2, b-1\}$, with $b-1$ visited. Strands. *Stem (D) yields exactly (W3) reversed, (W4) reversed, (W5) reversed.*

Collecting the stems: every admissible visit is one of (W1)–(W5) up to reversal, and the maximum size is 3. ☐

Remark B.4 (Degenerate and small cases)**.** For $b = 5$ the highway $D = \{3\}$ degenerates and Lemmas B.1–B.2 do not apply verbatim, but direct enumeration confirms that the classification's *shape* persists: $T_5' = (0, 1, 4, 3, 2)$ has admissible visits exactly (0), (0, 1, 2), (0, 1, 4), (2, 1, 4), (2, 3, 4), so $M(T_5') = 3$. For $b = 3, 4$ no member of either gadget family exists, and the exhaustive minima $\mu(3) = \mu(4) = 3$ are computed directly (Section 11). Note also the role pairs realized here — $\{p_0, v_1\}$ and $\{v_0, p_1\}$ appear, while T_b forbids exactly those two — so the two gadget families attain the transit floor through genuinely different visit structures.

Acknowledgments

This paper merges and supersedes the author's two earlier notes on this problem – an extremal-conjecture note containing the conversion lemma, the block-reversal family, and the $n = 18$ refutation, and a second note proving the $O(\sqrt{n})$ bound – and extends them with the transit-floor theorem (Section 9) and the block-bound refutations (Section 10), which were obtained in a follow-up research campaign on the open problems of the earlier version. The proofs and computations were AI-assisted by Claude, under the Periapsis research structure. Because of that, I have tried to make the paper checkable without trusting either the author or the assistant: the central case analyses are classifications (Theorems 7.4 and B.3) and elementary matching arguments (Theorem 9.6) whose every step is locally verifiable; every computational claim is reproducible from the ancillary artifact, which includes independent verifiers written directly from the definitions; and the main theorem has a second proof in Appendix A. This is an unrefereed preprint; no journal referee has yet checked it.

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